Project Report

On

Career Compass

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CERTIFICATE

This is to certify that Mr. **Tushar Thakur** , of BCA 6th Semester bearing Roll No. **8004** is a bonafide student of DAV College, Chandigarh (Batch 2025-2026) has completed the project entitled " **Career Compass** " under my supervision & Guidance. It is further certified that the work done in this project is a result of candidate's own efforts. I wish him all the success in his life.

Date:

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1. Introduction

1.1 Overview of the Project

In today's rapidly evolving world, choosing the right career path has become increasingly complex for students. With the availability of numerous career options and constant changes in industry demands, students often face confusion and uncertainty while making career decisions. Traditional career guidance systems rely heavily on manual counseling, which may not always be personalized, accessible, or scalable.

The **Career Compass** project is a web-based platform designed to assist students in making informed career decisions based on their academic performance, interests, and preferences. The system collects relevant data from users and processes it to provide suitable career recommendations.

This platform aims to bridge the gap between students and career guidance by offering a structured, data-driven approach. It integrates user input with predefined logic and decision-making mechanisms to generate personalized suggestions. Additionally, the system includes an admin interface that allows administrators to manage data, monitor users, and update career-related information dynamically.

The project is developed using modern web technologies, ensuring responsiveness, usability, and scalability. By providing an intuitive interface and accurate recommendations, Career Compass enhances the overall decision-making process for students.

Furthermore, the system emphasizes ease of use and accessibility, allowing students from different educational backgrounds to interact with the platform without requiring technical expertise. The interface is designed to be simple and intuitive, ensuring that users can easily navigate through different sections such as data input, recommendation results, and dashboard views.

Another important aspect of the project is its ability to store and manage large amounts of user data efficiently. By utilizing a structured database system, the platform ensures that user information is securely stored and can be retrieved quickly when needed. This also enables the system to maintain consistency in recommendations and improve performance over time.

In addition, the Career Compass platform supports automation in data processing, reducing manual effort and minimizing errors. The integration of automated workflows ensures that once the user submits their information, the system processes it seamlessly and generates results without delay. This improves the overall efficiency and reliability of the system.

Overall, the Career Compass project represents a modern approach to career guidance by combining technology, data management, and user-centric design. It not only simplifies the decision-making process for students but also provides a scalable solution that can be enhanced with advanced features in the future.

1.2 Purpose of the Project

The primary purpose of the Career Compass system is to provide a reliable and efficient solution for career guidance. Many students lack access to proper counseling resources, which leads to uninformed decisions that may affect their future.

This project aims to:

- Provide personalized career recommendations based on user input
- Reduce confusion among students regarding career choices
- Offer a centralized platform for career-related information
- Automate the career guidance process to make it faster and more accessible
- Assist administrators in managing and updating career data efficiently

The system is designed to be user-friendly and accessible to a wide range of users, ensuring that students from different backgrounds can benefit from it.

1.3 Scope of the Project

The scope of the Career Compass project includes the development and implementation of a web-based application that provides career recommendations to students.

The system covers the following functionalities:

- User registration and login system

- Input collection (academic details, interests, preferences)
- Processing of user data to generate career suggestions
- Display of recommended career paths
- Admin panel for managing system data
- Dynamic updates to career options and recommendations

The platform is intended primarily for students at the school and college level who are in the process of selecting their career paths. It can also be extended in the future to include advanced features such as AI-based recommendations, integration with job portals, and real-time analytics.

However, the current system is limited to predefined logic and does not incorporate advanced machine learning algorithms. It also assumes that users provide accurate input data for effective recommendations.

1.4 Existing System

The existing system for career guidance is largely manual and lacks personalization. Students typically rely on:

- Career counseling sessions
- Advice from teachers or parents
- Online articles and generic career websites

These methods have several limitations:

- Lack of personalization
- Limited accessibility to professional guidance
- Time-consuming and sometimes costly
- Information may be outdated or irrelevant
- No centralized system for managing user data

Additionally, many existing online platforms provide general recommendations without considering individual user profiles in depth. This often leads to confusion and incorrect career choices.

1.5 Proposed System

The proposed system, Career Compass, is designed to overcome the limitations of the existing system by providing a structured and automated approach to career guidance.

Key features of the proposed system include:

- Personalized career recommendations based on user input
- Interactive and user-friendly web interface
- Centralized database for storing user and career data
- Admin panel for efficient data management
- Faster and more accurate decision-making support

The system uses a rule-based approach to analyze user data and generate suitable career options. It ensures that recommendations are relevant and tailored to individual users.

By digitizing the career guidance process, the proposed system improves accessibility, reduces dependency on manual counseling, and enhances the overall efficiency of career decision-making.

Conclusion of Introduction:

The Career Compass system provides a modern solution to traditional career guidance challenges. By leveraging web technologies and structured data processing, it offers a reliable platform for students to explore and choose suitable career paths effectively.

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2. Objectives

The objectives of the Career Compass project define the goals that the system aims to achieve. These objectives focus on improving the career decision-making process for students by providing a structured, efficient, and user-friendly platform.

2.1 Primary Objective

The primary objective of the Career Compass system is to develop a web-based application that provides personalized career recommendations to students based on their academic performance, interests, and preferences.

The system aims to simplify the complex process of career selection by offering accurate and relevant suggestions. It ensures that students can make informed decisions without relying solely on external guidance or generic information sources.

2.2 Secondary Objectives

In addition to the primary goal, the project includes several secondary objectives that enhance the overall functionality and effectiveness of the system.

2.2.1 To Provide Personalized Career Guidance

One of the key objectives is to deliver career recommendations tailored to individual users. The system collects user-specific data and processes it to generate suggestions that align with their strengths and interests.

This personalized approach helps students identify suitable career paths more effectively compared to traditional methods.

2.2.2 To Develop a User-Friendly Interface

The system is designed with a focus on simplicity and ease of use. The objective is to ensure that users can interact with the platform without requiring technical knowledge.

A clean and intuitive interface allows users to:

- Easily enter their details
- Navigate through the system
- View recommendations without confusion

2.2.3 To Automate the Career Recommendation Process

Another important objective is to automate the process of career guidance. Instead of manual counseling, the system processes user data automatically and generates results instantly.

This reduces:

- Time consumption
- Human error
- Dependency on external guidance

2.2.4 To Provide Centralized Data Management

The system aims to maintain a centralized database that stores user information and career-related data efficiently.

This objective ensures:

- Easy retrieval of data
- Consistency in recommendations
- Better system organization

2.2.5 To Support Administrative Control

The project includes an admin panel that allows administrators to manage system data. This objective ensures that the platform remains updated and relevant.

Administrators can:

- Add or update career options
- Monitor user activity
- Maintain system performance

2.2.6 To Improve Decision-Making Efficiency

The system is designed to help students make quicker and more informed decisions regarding their careers.

By providing structured recommendations, the platform:

- Reduces confusion
- Increases clarity
- Enhances confidence in decision-making

2.2.7 To Ensure Accessibility and Scalability

The project aims to create a system that is accessible to a wide range of users and can be expanded in the future.

This includes:

- Access from different devices (desktop, mobile)
- Ability to add advanced features later
- Scalability for larger user bases

2.3 Long-Term Objectives

The Career Compass system is designed with future enhancements in mind. Some long-term objectives include:

- Integration of Artificial Intelligence for smarter recommendations
- Real-time career trend analysis
- Integration with job portals and educational platforms
- Personalized dashboards with analytics
- Continuous improvement based on user feedback

Conclusion of Objectives:

The objectives of the Career Compass project focus on creating an efficient, user-centric, and scalable solution for career guidance. By achieving these objectives, the system aims to transform the traditional approach to career decision-making into a more structured and technology-driven process.

3. Requirement Analysis

Requirement Analysis is a crucial phase in the development of any software system. It involves identifying, analyzing, and documenting the needs and expectations of users. This phase ensures that the system being developed meets user requirements effectively and functions as intended.

In the Career Compass project, requirement analysis focuses on understanding the challenges faced by students in choosing a career path and designing a system that addresses those challenges through structured and automated solutions. This involves studying user needs, identifying common difficulties such as lack of guidance and confusion among multiple career options, and translating these issues into clear system requirements. By doing so, the development process becomes more organized, ensuring that the final system effectively supports students in making informed and confident career decisions.

3.1 Problem Analysis

Choosing a career is one of the most critical decisions in a student's life. However, many students face difficulties due to lack of proper guidance, limited access to professional counseling, and overwhelming information available online.

The major problems identified are:

- Lack of personalized career guidance
- Dependence on manual counseling methods
- Limited awareness of available career options
- Difficulty in analyzing individual strengths and interests
- Time-consuming and sometimes costly guidance processes

Students often rely on general advice, which may not align with their personal abilities or aspirations. This leads to confusion, poor decision-making, and dissatisfaction in the long run.

The absence of a centralized and automated system further complicates the process. Therefore, there is a need for a solution that provides accurate, personalized, and easily accessible career recommendations.

3.2 Software Requirement Specification (SRS)

The Software Requirement Specification (SRS) is a formal document that describes the functional and non-functional requirements of the system. It acts as a blueprint for the development process.

3.2.1 Introduction to SRS

The SRS document defines the scope, objectives, and overall structure of the Career Compass system. It provides a detailed description of system functionalities, user interactions, and constraints.

The purpose of this SRS is to ensure that all stakeholders, including developers and users, have a clear understanding of the system requirements.

3.2.2 General Description

The Career Compass system is a web-based application designed to assist students in making informed career choices.

The system consists of two main components:

- User Module (Student Interface)
- Admin Module (Administrative Interface)

The user module allows students to input their details and receive career recommendations, while the admin module enables management of system data.

3.2.3 Product Perspective

The Career Compass system is a standalone application but can also be integrated with other systems in the future.

It interacts with:

- A database for storing user and career data
- Web browsers for user interaction

The system follows a client-server architecture where the frontend interacts with the backend and database to process user requests.

3.2.4 Product Functions

The major functions of the system include:

- User registration and authentication
- Collection of user data (academic details, interests)
- Processing and analysis of user input
- Generation of career recommendations
- Display of results to the user
- Admin management of data and system updates

Each function is designed to ensure smooth interaction and accurate output.

3.2.5 User Characteristics

The system is designed for different types of users:

Students:

- Basic computer knowledge
- Seeking career guidance
- Prefer simple and interactive interfaces

Administrators:

- Familiar with system management
- Responsible for maintaining data
- Require access to control features

The interface is designed to accommodate users with minimal technical expertise.

3.2.6 Constraints

The system operates under certain constraints:

- Requires internet connectivity
- Dependent on accurate user input
- Limited to predefined logic (no advanced AI in current version)

- Performance depends on server and database efficiency

These constraints are considered during system design and implementation.

3.2.7 Assumptions and Dependencies

The system assumes:

- Users provide correct and complete information
- The database is properly maintained
- The system is accessed through supported web browsers

Dependencies include:

- Web technologies (HTML, CSS, JavaScript)
- Database systems
- Hosting environment

3.3 Specific Requirements

3.3.1 External Interface Requirements

The system interacts with users through a graphical user interface (GUI).

User Interface:

- Simple and responsive design
- Forms for data input
- Dashboard for displaying results

Hardware Interface:

- Compatible with desktops, laptops, and mobile devices

Software Interface:

- Web browsers (Chrome, Edge, etc.)
- Database management system

3.3.2 Functional Requirements

Functional requirements define what the system should do.

Key functional requirements include:

- The system must allow user registration and login
- The system must collect and store user data
- The system must analyze user input
- The system must generate career recommendations
- The system must display results clearly
- The admin must be able to manage system data

Each function is essential for the proper working of the system.

3.3.3 Non-Functional Requirements

Non-functional requirements define system quality attributes.

- **Performance:** Fast response time
- **Usability:** Easy to use interface
- **Reliability:** Accurate recommendations
- **Security:** Protection of user data
- **Scalability:** Ability to handle more users

These ensure that the system performs efficiently under different conditions.

3.3.4 Hardware Requirements

The system requires minimal hardware:

- Computer or mobile device
- Internet connection
- Basic input/output devices

3.3.5 Software Requirements

The system requires:

- Web browser
- Frontend technologies (HTML, CSS, JavaScript)

- Backend (if applicable)
- Database system

Conclusion of Requirement Analysis:

The Requirement Analysis phase ensures that the Career Compass system is designed based on clear and well-defined user needs. By identifying functional and non-functional requirements, the system provides a strong foundation for development, ensuring efficiency, usability, and reliability.

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4. Software Design

Software Design is a crucial phase in system development where the overall architecture, components, modules, and data flow of the system are defined. It acts as a blueprint for the implementation phase and ensures that all system requirements are translated into a structured design.

In the Career Compass project, software design focuses on creating an efficient, scalable, and user-friendly system that can process user data and generate accurate career recommendations. The design phase ensures that all functional and non-functional requirements identified earlier are properly translated into a structured framework that guides the development process.

The system is designed with a strong emphasis on usability and performance so that users can easily interact with the platform without requiring advanced technical knowledge. At the same time, the backend is structured to handle data efficiently, ensuring quick processing and reliable output. Special attention is given to modularity, allowing different components of the system to function independently while still maintaining seamless integration.

Additionally, the design incorporates flexibility to allow future enhancements such as integration with advanced analytics or artificial intelligence-based recommendation systems. This ensures that the system remains adaptable to evolving technological trends and user needs.

This complete flow of operations is illustrated in **figure 4.1.1**, which represents the step-by-step working of the system through a flowchart.

4.1.1 Overall System Flow

The working of the system can be summarized in the following steps:

1. User registers/logs into the system
2. User inputs academic and personal details
3. System processes the input data
4. Career recommendations are generated
5. Results are displayed to the user
6. Admin manages system data

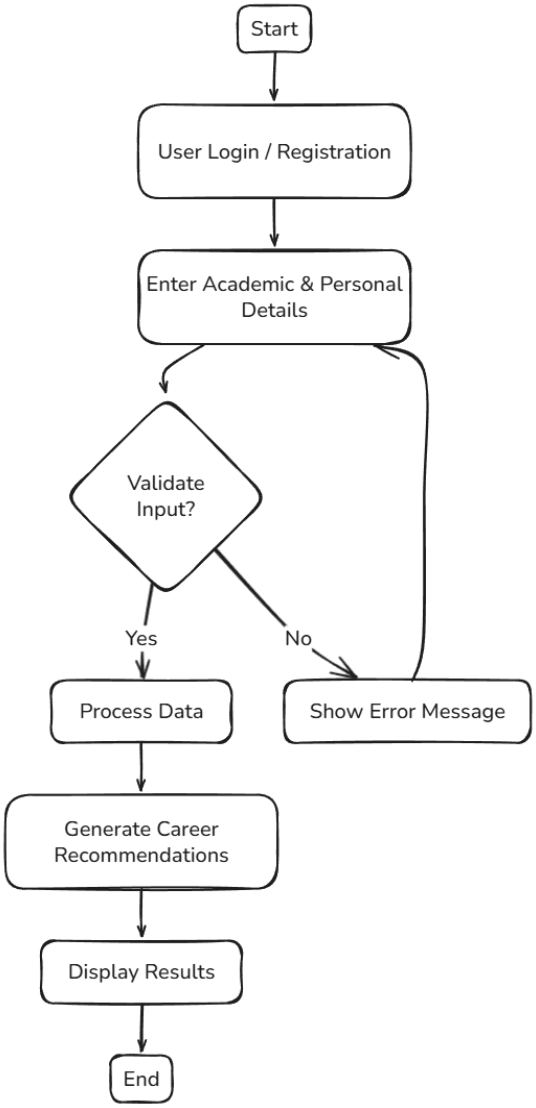


Figure: 4.1.1 – System Flow Diagram

4.2 Architectural Design

Architectural design defines the structure of the system, including components and their interactions.

The Career Compass system follows a **client-server architecture**.

4.2.1 Architecture Overview

The Career Compass system follows a modern cloud-based architecture using automation tools instead of a traditional backend server. The system consists of three main layers:

- **Presentation Layer (Frontend):** Handles user interaction through web pages.

- **Application Layer (Automation Logic):** Implemented using Make.com, which processes user input, applies logic, and generates recommendations.
- **Data Layer (Cloud Database):** Google Sheets is used as a database to store user information, academic data, and career recommendations.

This architecture eliminates the need for a dedicated backend server and simplifies system development while maintaining efficiency and scalability.

4.2.2 Architecture Diagram

The system architecture of Career Compass is based on a cloud-integrated model where the frontend interacts with automation services instead of a traditional backend. User requests from the frontend are processed through Make.com, which acts as the application logic layer. The processed data is stored and retrieved from Google Sheets, which serves as the system database.

This interaction is illustrated in **figure 4.2.1**, showing how data flows between the user interface, automation platform, and cloud database.



Figure: 4.2.1 – System Architecture Diagram

4.2.3 Advantages of Architecture

- Clear separation of components
- Easy maintenance and scalability
- Efficient data processing
- Improved system performance

4.3 Database Design

Database design defines how data is stored, organized, and managed within the system.

The Career Compass system uses **Google Sheets as a cloud-based database** to store and manage data. It provides a simple and efficient way to handle user information, academic records, and career recommendations without requiring a traditional database management system.

- User details
- Academic data
- Career options
- Recommendations

4.3.1 Entity Relationship (ER) Diagram

The Entity Relationship (ER) diagram represents the structure of the database and the relationships between different entities such as users, careers, and recommendations. It helps in understanding how data is organized and interconnected within the system.

The ER diagram of the Career Compass system is shown in **figure 4.3.1**, which highlights the entities and their relationships.

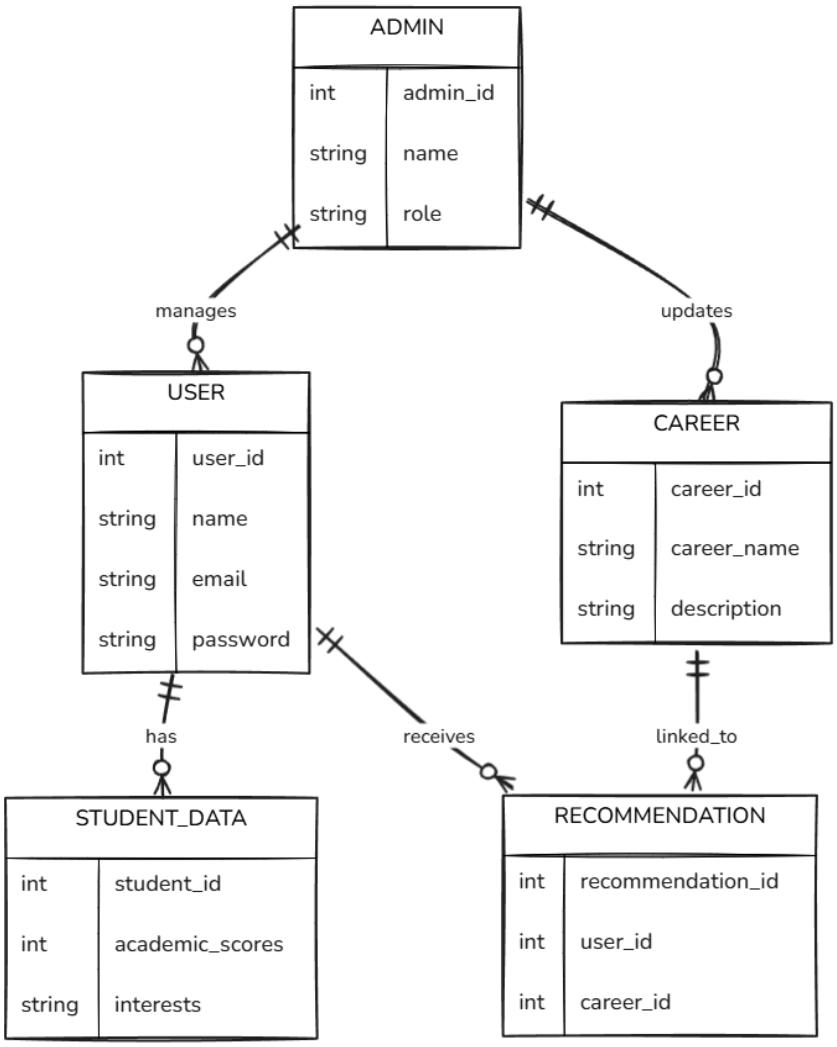


Figure: 4.3.1 – ER Diagram

4.3.2 Database Tables

1. User Table

- User_ID (Primary Key)
- Name
- Email
- Password

2. Student Data Table

- Student_ID
- Academic Scores
- Interests

3. Career Table

- Career_ID
- Career_Name
- Description

4. Recommendation Table

- Recommendation_ID
- User_ID
- Career_ID

4.3.3 Data Relationships

- One user can have multiple recommendations
- Each recommendation links to a career
- Admin manages all data

4.4 User Interface Design

User Interface (UI) design focuses on creating an intuitive and user-friendly interaction between the user and the system.

The Career Compass system provides a clean and responsive interface.

4.4.1 UI Components

- Login Page
- Registration Page
- Dashboard
- Input Form
- Recommendation Page
- Admin Panel

4.4.2 User Interaction Flow

This section describes how users interact with the Career Compass system from start to finish. The process begins with user authentication, where users either log in or register. Once authenticated, users are guided to input their academic details and personal interests.

The system then processes this information using the automation logic implemented through Make.com. Based on the processed data, relevant career recommendations are generated and displayed to the user in an organized format.

Additionally, administrators can access the system to manage user data and update career-related information. This structured interaction ensures a smooth and efficient user experience throughout the system.

4.4.3 UI Design Principles

- Simplicity
- Consistency
- Responsiveness
- Accessibility

The interface ensures that users can easily navigate and interact with the system.

4.5 Component-Level Design

The processing module is implemented using Make.com, which automates data processing and recommendation generation, while Google Sheets acts as the data storage component.

4.5.1 User Module

Handles:

- Registration
- Login
- Data input
- Viewing recommendations

4.5.2 Admin Module

Handles:

- Managing users
- Updating career data
- Monitoring system activity

4.5.3 Processing Module

Responsible for:

- Analyzing user input
- Applying decision logic
- Generating recommendations

4.5.4 Component Diagram

The component diagram represents the internal structure of the system by showing different modules and their interactions. It helps in understanding how various components such as user module, admin module, and processing module work together.

The component-level design is illustrated in **figure 4.5.1**, which provides a clear view of system modularization.

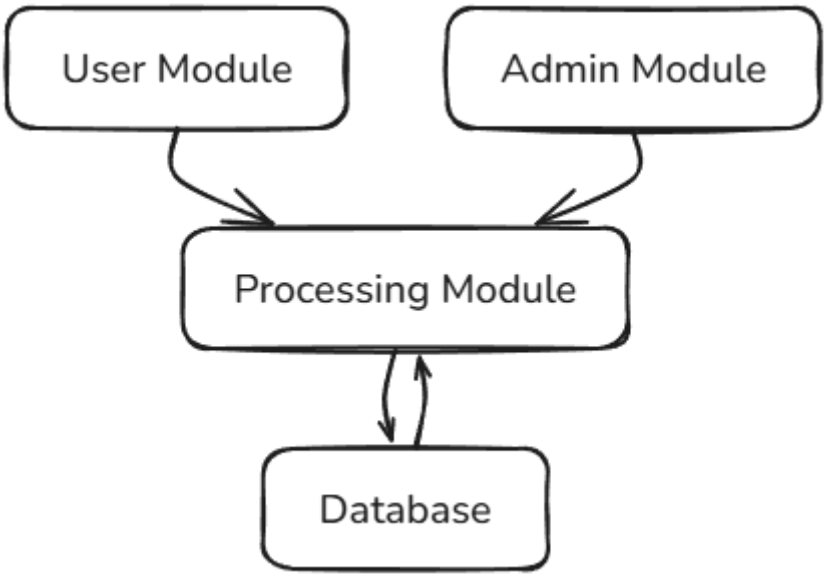


Figure: 4.5.1 – Component Diagram

4.6 Data Flow Design

This section provides a detailed explanation of how data flows through the system, outlining each stage from initial user input to the final output. It describes how user-provided information is collected through the frontend interface, transmitted to the Make.com automation platform for processing, and then stored or retrieved from Google Sheets, which serves as the system's cloud-based database. Additionally, it highlights how the processed data is used to generate meaningful career recommendations, which are then sent back to the user interface for display. This flow ensures efficient data handling, real-time processing, and seamless interaction between all components of the system.

4.6.1 Level 0 DFD

The Level 0 Data Flow Diagram (Context Diagram) provides a high-level view of the system, showing how external entities interact with the system. It represents the system as a single process along with its inputs and outputs.

The Level 0 DFD of the Career Compass system is shown in **figure 4.6.1.**

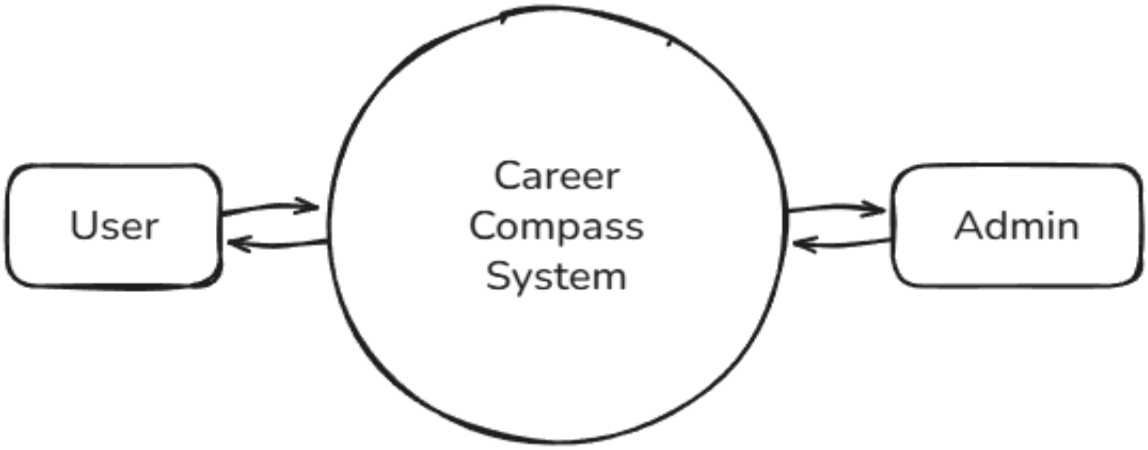


Figure: 4.6.1 – Level 0 DFD

4.6.2 Level 1 DFD

The Level 1 Data Flow Diagram provides a more detailed view of the system by breaking down the main process into sub-processes. It shows how data flows between different modules within the system.

This detailed representation is illustrated in **figure 4.6.2.**

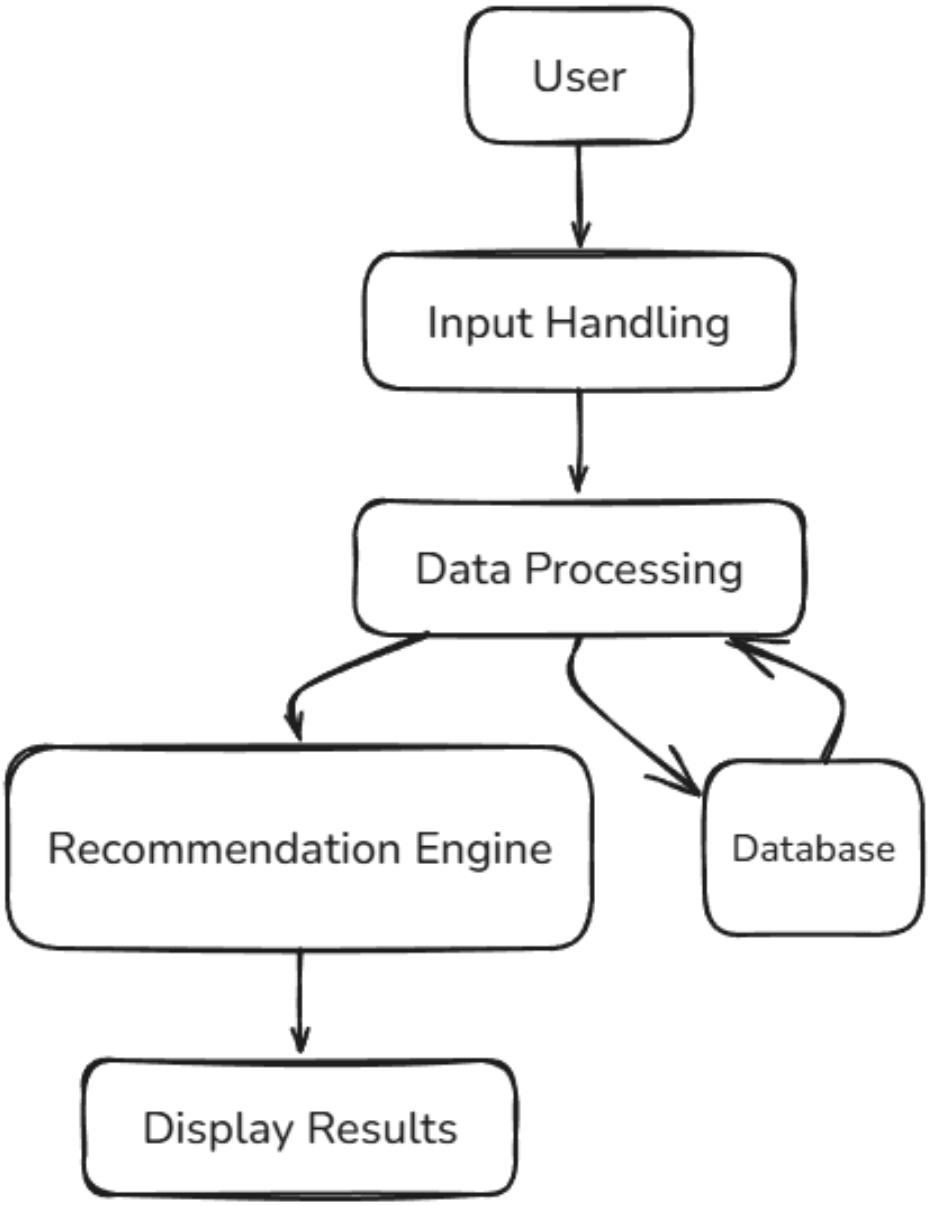


Figure: 4.6.2 – Level 1 DFD

4.6.3 Level 2 DFD

The Level 2 Data Flow Diagram further expands the internal processes of the system, providing a deeper understanding of data movement and processing logic.

The detailed breakdown is shown in **figure 4.6.3.**

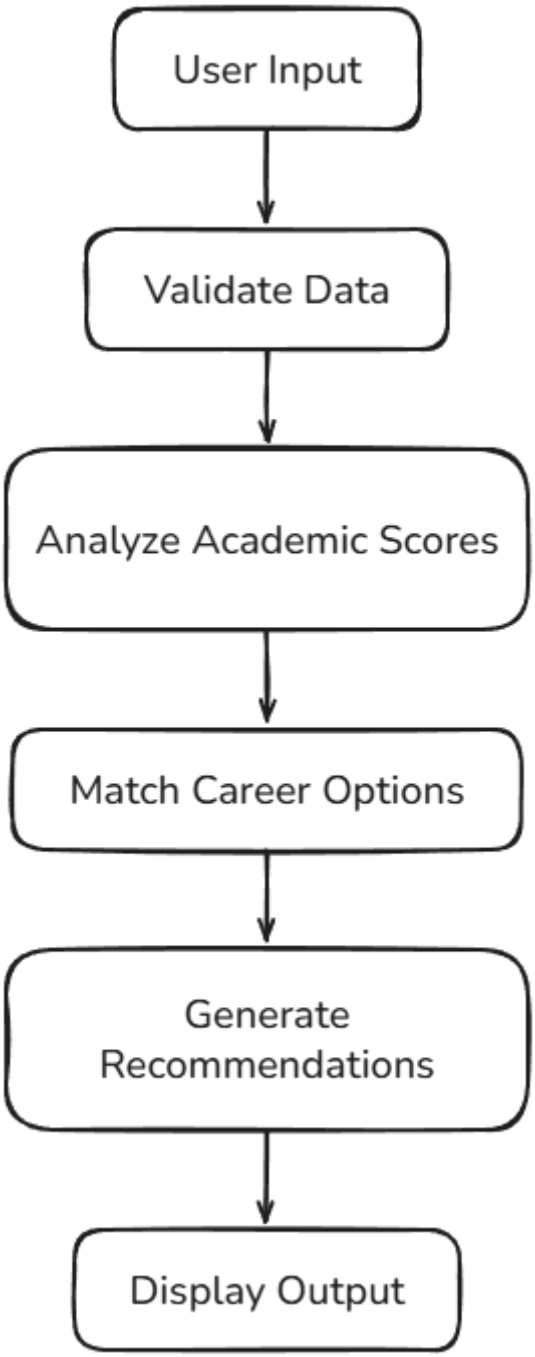


Figure: 4.6.3 – Level 2 DFD

4.7 Design Constraints

- Internet dependency
- Limited to rule-based logic
- Browser compatibility
- Database dependency

Conclusion of Software Design:

The Software Design phase defines the structure and behavior of the Career Compass system. By incorporating architectural design, database design, UI design, and data flow diagrams, the system ensures efficient implementation and smooth operation. The inclusion of diagrams and structured design enhances clarity and improves the overall quality of the project documentation.

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5. Coding and Implementation

The Coding and Implementation phase is the stage where the system design is transformed into a fully functional and operational application. In this phase, all the modules, components, and functionalities defined during the design stage are carefully developed, tested, and integrated to create the final system. This phase plays a crucial role in ensuring that the theoretical design is successfully converted into a practical solution that meets user requirements.

The Career Compass system is implemented as a web-based application using modern technologies that focus on simplicity, efficiency, and scalability. Instead of relying on a traditional backend server, the system utilizes Make.com as an automation platform to handle the core application logic. Make.com acts as an intermediary that processes user inputs, applies decision-making logic, and manages communication between different components of the system.

For data storage and management, Google Sheets is used as a cloud-based database. It stores user information, academic details, and generated career recommendations in a structured format. This approach eliminates the need for complex database management systems while still providing reliable and accessible data storage.

The integration between the frontend, Make.com, and Google Sheets ensures smooth data flow throughout the system. When a user submits information through the frontend interface, the data is sent to Make.com via webhooks. Make.com processes the data, applies predefined rules or logic, and stores the results in Google Sheets. The processed output is then returned to the user in the form of career recommendations.

This approach simplifies development, reduces infrastructure complexity, and enables seamless integration between different system components. Additionally, it enhances flexibility, as changes in logic or data handling can be easily implemented within Make.com without modifying the entire system. Overall, the implementation strategy ensures that the Career Compass system is efficient, scalable, and easy to maintain.

5.1 Programming Approach

The Career Compass system follows a modular, event-driven, and API-based programming approach to ensure flexibility, scalability, and ease of maintenance. This approach allows different components of the system to function independently while still being interconnected through well-defined communication channels.

Key Aspects:

- **Modular Design:**
  The system is divided into independent modules such as user authentication (login and registration), data input, data processing, and career recommendation. Each module is designed to perform a specific function, making it easier to develop, test, and maintain the system. This modular structure also allows future enhancements without affecting the entire system.
- **Event-Driven Execution:**
  The system operates based on user-triggered events. For example, when a user submits a form, an event is triggered that initiates data validation and sends the data to the backend (Make.com). This approach improves responsiveness and ensures that actions are performed only when required.
- **API-Based Communication:**
  Communication between the frontend and backend is handled using HTTP requests through Webhooks. The frontend sends user data in JSON format to Make.com, which processes the data and returns the required output. This ensures seamless integration between different components.
- **No Traditional Backend:**
  Instead of using conventional backend technologies such as Node.js, PHP, or Python frameworks, the system uses Make.com as an automation platform. This reduces development complexity and eliminates the need for server management while still providing powerful data processing capabilities.

5.2 Technologies Used

The development of the Career Compass system involves a combination of frontend technologies and cloud-based automation tools.

5.2.1 Frontend Technologies

The frontend is responsible for user interaction and interface design.

- **HTML (HyperText Markup Language):**
  Used to structure the web pages and define elements such as forms, buttons, input fields, headings, and layouts. It provides the basic skeleton of the application and ensures that all content is properly organized and accessible to users.
- **CSS (Cascading Style Sheets):**
  Used to style the interface, improve visual appearance, and ensure responsiveness across different devices. CSS helps in designing layouts, applying colors, fonts, spacing, and making the user interface visually appealing and user-friendly.
- **JavaScript:**
  Used to add interactivity and dynamic behavior to the web pages. It is responsible for validating user input, handling form submissions, and enabling communication with external services such as Make.com through webhooks. JavaScript ensures smooth user experience by providing real-time feedback and processing without reloading the page.

5.2.2 Backend (Automation Logic)

Instead of using a traditional backend server such as Node.js or PHP, the Career Compass system utilizes **Make.com** as an automation-based backend solution. Make.com acts as the central processing unit of the system, handling all data operations and logical processing without requiring manual server-side coding.

Make.com works through automated workflows known as scenarios, which are triggered when data is received from the frontend via webhooks. Once triggered, the platform performs a sequence of predefined actions to process the data efficiently.

- **Make.com:**
  Acts as the core processing engine of the system. Its functionalities include:
    - Receiving user data from the frontend through webhooks
    - Validating and formatting the input data
    - Applying rule-based logic to analyze academic performance and interests
    - Generating suitable career recommendations based on predefined conditions
    - Sending processed data to Google Sheets for storage
    - Returning responses back to the frontend if required

This approach eliminates the need for maintaining a dedicated backend server, reduces development complexity, and ensures faster deployment and scalability.

5.2.3 Database

- **Google Sheets:**
  Google Sheets is used as a cloud-based database for storing and managing all system data. It provides a simple yet effective way to handle structured data without requiring a traditional database management system.

The data stored in Google Sheets includes:

- User details (Name, Email, etc.)
- Academic data (Marks, Subjects, etc.)
- User interests and preferences
- Career options and recommendation results

Each row in the sheet represents a unique user entry, while columns define different attributes of the data. Make.com interacts directly with Google Sheets to insert, update, and retrieve data as needed.

Using Google Sheets as a database offers several advantages:

- Easy accessibility from anywhere
- Real-time data updates
- Seamless integration with Make.com
- No requirement for complex database setup or maintenance

This cloud-based approach ensures efficient data management while keeping the system lightweight and easy to maintain.

5.2.4 Tools & Environment

- **Web Browser (Chrome/Edge):** Used for testing and running the application
- **Code Editor (VS Code):** Used for writing and editing code
- **Internet Connectivity:** Required for accessing cloud services like Make.com and Google Sheets

---

5.3 Source Code Implementation

The system is implemented in a modular way, ensuring that each component performs a specific function while maintaining smooth integration.

The workflow of the system is as follows:

1. User interacts with the frontend interface
2. Data is collected through input forms
3. Data is sent to Make.com using webhooks
4. Make.com processes the data
5. Processed data is stored in Google Sheets
6. Career recommendations are generated
7. Results are sent back and displayed to the user

5.3.1 Implementation Flow Diagram

The implementation flow diagram illustrates the step-by-step process through which data moves within the Career Compass system. It provides a clear understanding of how user input is collected, processed, stored, and transformed into meaningful career recommendations. The diagram highlights the interaction between the frontend interface, Make.com automation platform, and Google Sheets database, ensuring transparency in system operations and data flow.



Figure: 5.3.1 – Implementation Flow Diagram

5.3.2 Frontend Implementation

The frontend of the Career Compass system is designed to provide a simple, intuitive, and user-friendly interface that allows users to interact with the system efficiently. It serves as the primary point of interaction between the user and the application, ensuring that users can easily input their data and view the generated career recommendations.

The frontend is developed using HTML, CSS, and JavaScript, which together help in creating a responsive and visually appealing interface. HTML is used to structure the content of the web pages, CSS is applied to enhance the visual design and layout, and JavaScript is used to add interactivity and handle user inputs dynamically.

Special attention is given to usability and accessibility, ensuring that users with minimal technical knowledge can navigate the system without difficulty. The interface includes clearly labeled input fields, buttons, and instructions to guide users through the process of entering their academic details and interests.

Additionally, the frontend is designed to validate user inputs before sending data to the backend (Make.com), reducing errors and ensuring that only correct and complete information is processed. This improves the overall efficiency and reliability of the system.

Overall, the frontend implementation plays a crucial role in enhancing user experience and ensuring smooth communication between the user and the system.

5.4 Source Code Snippets

This section presents **detailed and extended source code** of the Career Compass system. The code is organized into three layers:

- HTML (Structure)
- CSS (Design)
- JavaScript (Functionality + Make.com Integration)

Each layer includes the following modules:

- Main Website
- Admin Page
- Student Page
- Assessment Page

5.4.1 HTML Structure (Detailed)

(A) Main Website

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Career Compass</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>

<header class="main-header">

  <h1>Career Compass</h1>
```

```
  <nav>
    <ul>
      <li><a href="#home">Home</a></li>
      <li><a href="#student">Student</a></li>
      <li><a href="#assessment">Assessment</a></li>
      <li><a href="#admin">Admin</a></li>
    </ul>
  </nav>
</header>
<section id="home" class="hero">
  <h2>AI-Based Career Guidance</h2>
  <p>Analyze your skills and get personalized recommendations.</p>
  <button onclick="scrollToSection('student')">Get Started</button>
</section>
<footer>
  <p>© 2026 Career Compass</p>
</footer>
</body>
</html>
```

(B) Admin Page

```
<div class="admin-dashboard">
  <h2>Admin Dashboard</h2>
  <div class="admin-controls">
    <input type="text" id="searchInput" placeholder="Search student">
    <button onclick="searchUser()">Search</button>
    <button onclick="refreshData()">Refresh</button>
  </div>
  <table>
    <thead>
      <tr>
        <th>Name</th>
        <th>Email</th>
        <th>Marks</th>
        <th>Interest</th>
        <th>Recommendation</th>
      </tr>
    </thead>
    <tbody id="adminTable"></tbody>
  </table>
</div>
```

(C) Student Page

```
<form id="studentForm">
  <h2>Student Registration</h2>

  <input type="text" id="name" placeholder="Full Name" required>
  <input type="email" id="email" placeholder="Email Address" required>
  <input type="tel" id="phone" placeholder="Phone Number">
  <input type="text" id="city" placeholder="City">

  <select id="education">
    <option value="">Select Education</option>
    <option>10th</option>
    <option>12th</option>
    <option>Graduate</option>
  </select>

  <button type="submit">Save & Continue</button>
</form>
```

(D) Assessment Page

```
<form id="assessmentForm">
  <h2>Career Assessment</h2>
  <input type="number" id="marks" placeholder="Marks (%)" required>
  <select id="interest">
    <option value="">Select Interest</option>
    <option value="science">Science</option>
    <option value="commerce">Commerce</option>
    <option value="arts">Arts</option>
  </select>
  <select id="skills">
    <option value="">Skill Level</option>
    <option>Beginner</option>
    <option>Intermediate</option>
    <option>Advanced</option>
  </select>
  <textarea id="comments" placeholder="Additional Info"></textarea>
  <button type="submit">Generate Recommendation</button>
</form>
<div id="resultBox"></div>
```

5.4.2 CSS Styling (Extended)

(A) Main Website

```
body { font-family: Arial; margin:0; background:#f5f7fa; }

.main-header { display:flex; justify-content:space-between; padding:15px; background:#222; color:#fff; }

nav ul { display:flex; list-style:none; }

nav li { margin-right:20px; }

nav a { color:white; text-decoration:none; }

.hero { text-align:center; padding:60px; }

.hero button { padding:10px 20px; background:#007bff; color:white; border:none; }

footer { text-align:center; padding:10px; background:#ddd; }
```

(B) Admin Page

```
.admin-dashboard { padding:20px; background:#eef2ff; border-radius:10px; }

.admin-controls { margin-bottom:15px; }

.admin-controls input { padding:8px; }

table { width:100%; border-collapse:collapse; }

th, td { border:1px solid #ccc; padding:10px; text-align:center; }

tr:hover { background:#f1f1f1; }
```

(C) Student Page

```
#studentForm {
  width:400px; margin:auto; padding:25px;
  background:white; border-radius:12px;
  box-shadow:0 0 10px rgba(0,0,0,0.1);
}
#studentForm input, select {
  width:100%; margin:10px 0; padding:10px;}
button { cursor:pointer; }
```

(D) Assessment Page

```
#assessmentForm {
  width:400px; margin:auto; padding:25px;
  background:#fafafa;
}
#assessmentForm select, textarea, button {
  width:100%; margin-top:10px; padding:10px;
}
#resultBox {
  margin-top:20px; padding:15px;
  background:#d4edda; border-radius:8px;
}
```

5.4.3 JavaScript Functionality (with Make.com Integration)

(A) Main Website

```
function scrollToSection(id) {
  document.getElementById(id).scrollIntoView({ behavior: "smooth" });
}
```

(B) Admin Page

```
function loadAdminData() {
  console.log("Fetching data from Google Sheets via Make.com...");
}

function searchUser() {
  let query = document.getElementById("searchInput").value;
  console.log("Searching:", query);
}

function refreshData() {
  loadAdminData();
}
```

(C) Student Page (with Make.com Webhook)

```
document.getElementById("studentForm").addEventListener("submit", function(e) {
  e.preventDefault();

  let data = {
    name: document.getElementById("name").value,
    email: document.getElementById("email").value,
    phone: document.getElementById("phone").value,
    city: document.getElementById("city").value,
    education: document.getElementById("education").value
  };

  fetch("https://hook.make.com/your-webhook-url", {
    method: "POST",
    headers: { "Content-Type": "application/json" },
    body: JSON.stringify(data)
  })
  .then(res => res.json())
  .then(() => alert("Student data saved successfully"))
  .catch(() => alert("Error sending data"));
});
```

(D) Assessment Page (Recommendation + Make.com)

```
document.getElementById("assessmentForm").addEventListener("submit", function(e) {
  e.preventDefault();

  let marks = document.getElementById("marks").value;
  let interest = document.getElementById("interest").value;

  let recommendation = getRecommendation(marks, interest);

  document.getElementById("resultBox").innerHTML =
    "Recommended Career: " + recommendation;

  let data = { marks, interest, recommendation };

  fetch("https://hook.make.com/your-webhook-url", {
    method: "POST",
    headers: { "Content-Type": "application/json" },
    body: JSON.stringify(data)
  })
  .then(() => console.log("Assessment sent"));
});
```

```
function getRecommendation(marks, interest) {
  if (marks >= 85 && interest === "science") return "Engineering";
  if (interest === "commerce") return "Business";
  return "General Degree";
}
```

Conclusion

The above expanded code demonstrates a complete working structure of the Career Compass system. It integrates frontend design, user interaction, validation, and backend automation using Make.com and Google Sheets, ensuring a scalable and efficient architecture.

5.5 Make.com Integration

Make.com serves as the core automation and processing engine of the Career Compass system. Instead of relying on a traditional backend server, Make.com is used to handle all application logic, data processing, and communication between the frontend and the database (Google Sheets). This approach enables seamless integration, reduces development complexity, and allows real-time data handling.

Make.com operates using scenarios, which are workflows consisting of multiple modules connected in a sequence. Each module performs a specific task such as receiving data, processing it, storing it, or sending a response.

5.5.1 Workflow Steps

The workflow implemented in Make.com follows a structured sequence of operations:

1. **Receive Data via Webhook:**
   The process begins when the frontend sends user input data (such as marks and

interests) to a Make.com webhook URL. This webhook acts as the entry point for the automation workflow.

2. **Data Parsing and Validation:**

   The received data is parsed into structured fields. Validation checks are performed to ensure that all required inputs are present and correctly formatted.

3. **Data Processing:**

   The system analyzes the academic scores and user interests. Conditional logic is applied to categorize the data and prepare it for recommendation generation.

4. **Apply Recommendation Logic:**

   Based on predefined rules, the system determines suitable career options. For example, high marks in science subjects may lead to engineering or medical recommendations.

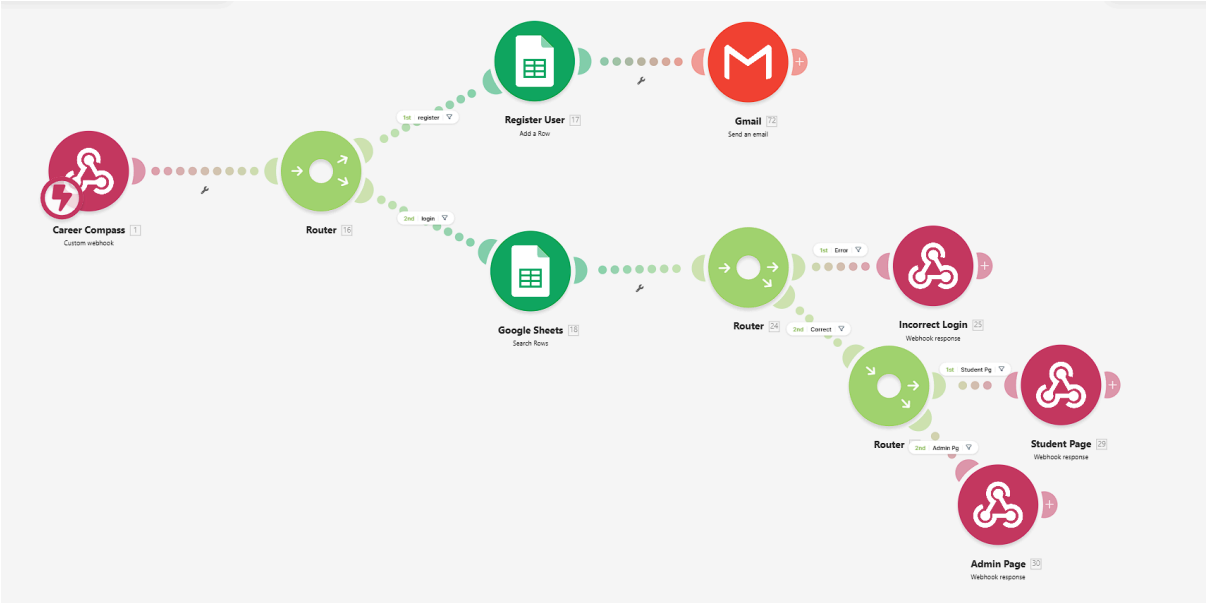
5. **Store Data in Google Sheets:**

   The processed data, along with the generated recommendation, is stored in Google Sheets. Each user entry is recorded as a new row, ensuring proper data organization.

6. **Send Response Back to Frontend:**

   After processing, the system sends a response back to the frontend. This response may include confirmation of submission or the recommended career path.

5.5.2 Make.com Scenario Diagram



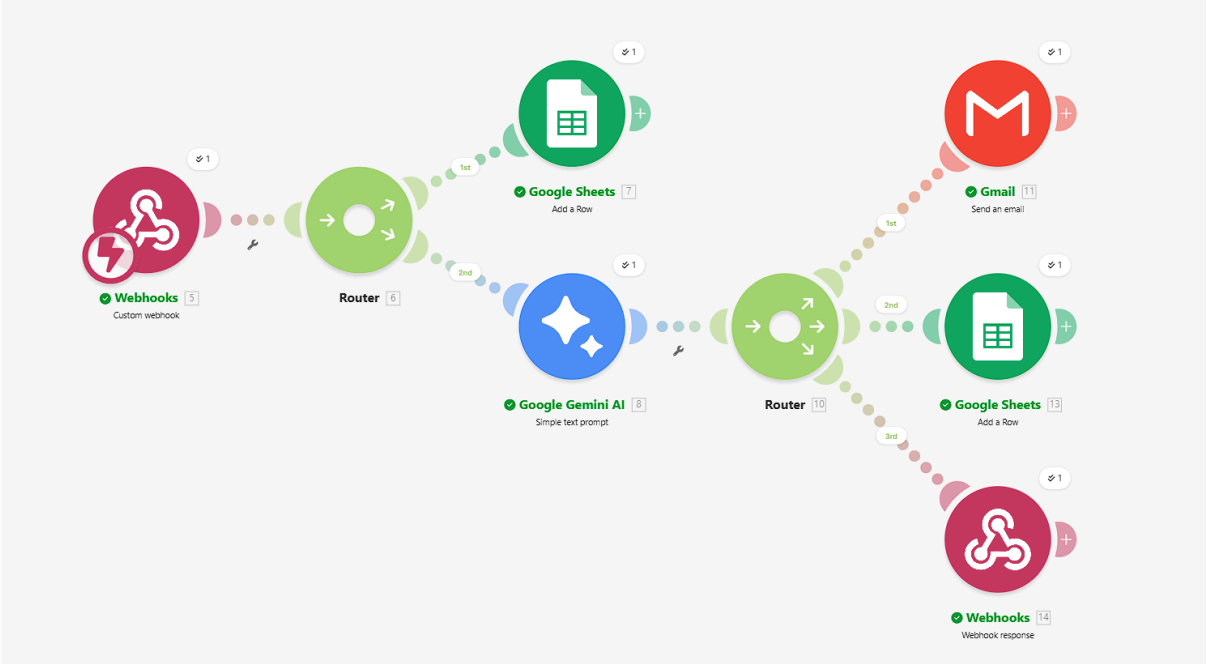


Figure: 5.5.1 – Make.com Workflow Scenario

The Make.com scenario diagram visually represents the sequence of modules used in the automation workflow. It typically includes:

- **Webhook Module:** Receives incoming data from the frontend
- **Tools/Functions Module:** Processes and transforms data
- **Router (Optional):** Directs data flow based on conditions
- **Google Sheets Module:** Stores processed data
- **Response Module:** Sends output back to the user

The diagram helps in understanding how data flows through the system and how different modules interact to produce the final output.

5.6 Google Sheets Integration

Google Sheets is used as the primary data storage system for the Career Compass application. Instead of relying on traditional database management systems, Google Sheets provides a cloud-based, easily accessible, and flexible solution for storing and managing user data. It allows real-time updates, seamless integration with Make.com, and simplifies data handling without requiring complex database configurations.

Google Sheets acts as a centralized repository where all user inputs, processed data, and generated recommendations are stored in a structured tabular format. This approach ensures that data can be easily viewed, edited, and maintained by administrators when required.

5.6.1 Data Structure

The data in Google Sheets is organized in rows and columns, where each row represents a unique user entry and each column represents a specific attribute of the user or system output.

The structure of the sheet is designed to ensure clarity, consistency, and ease of data processing. The following columns are maintained:

- **User ID:** A unique identifier assigned to each user entry for tracking purposes
- **Name:** Full name of the user
- **Email:** User's email address used for identification and communication
- **Marks:** Academic score entered by the user
- **Interests:** User's preferred field or area of interest (e.g., Science, Commerce, Arts)
- **Recommended Career:** Career option generated by the system based on input data
- **Timestamp:** Date and time when the data was submitted

This structured format ensures that data can be efficiently processed by Make.com and easily retrieved for analysis or reporting.

5.6.2 Data Storage Process

The process of storing data in Google Sheets involves multiple automated steps handled through Make.com integration:

1. **Data Submission:**
   The user enters their details through the frontend interface and submits the form.
2. **Webhook Trigger:**
   The submitted data is sent to Make.com via a webhook in JSON format.
3. **Data Processing:**
   Make.com receives the data and performs validation checks to ensure correctness and completeness.
4. **Data Mapping:**
   The incoming data fields are mapped to corresponding columns in Google Sheets.

5. **Row Creation:**

   A new row is automatically created in the Google Sheet, and the user data is inserted into the respective columns.

6. **Recommendation Storage:**

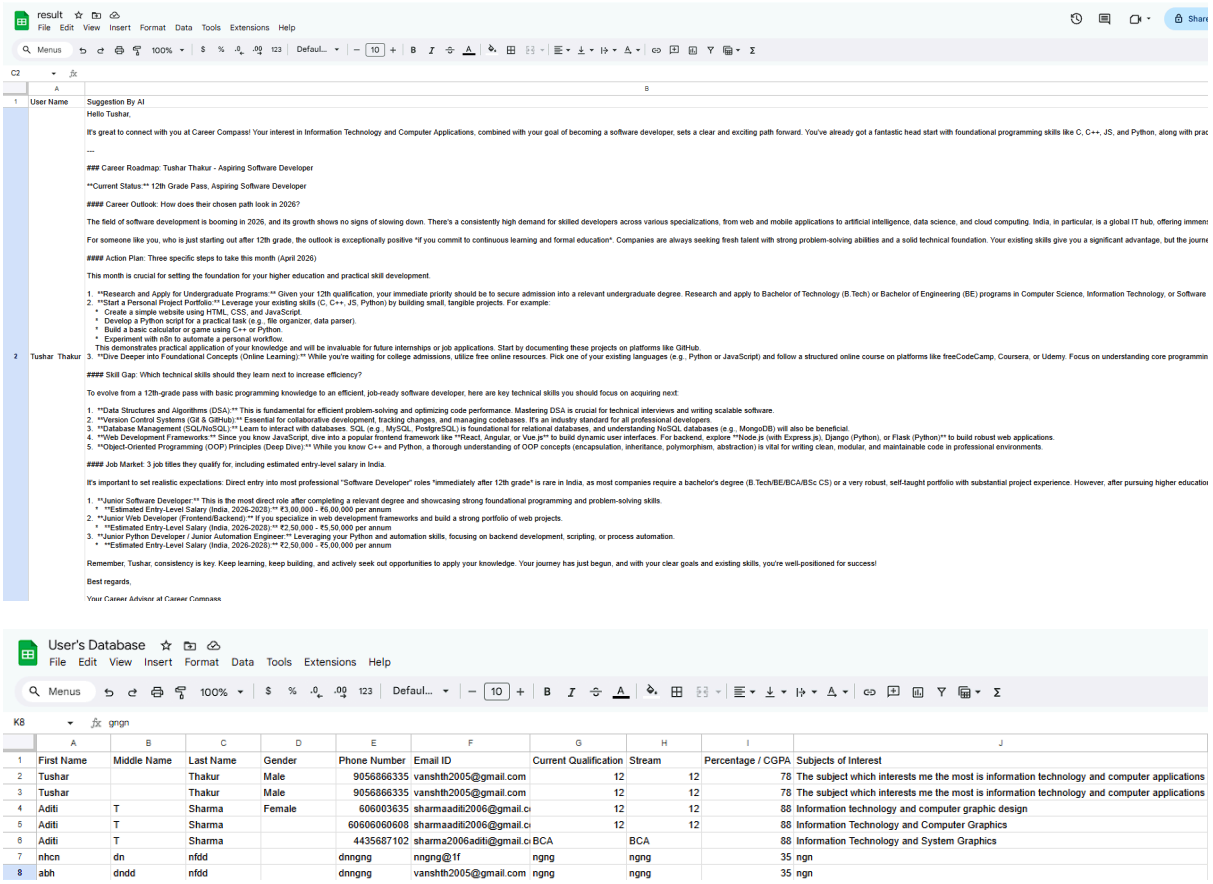
   After processing, the generated career recommendation is also stored in the same row.

7. **Data Retrieval:**

   Whenever required, Make.com can retrieve stored data from Google Sheets for further processing or display.

This automated workflow ensures accuracy, reduces manual effort, and maintains consistency in data storage.

	A	B	C	D	E
1	**Name**	**Email**	**Marks**	**Interests**	**Recommended Career**
2	Rahul Sharma	rahul@gmail.com	88	Science	Engineering
3	Priya Mehta	priya@gmail.com	76	Commerce	Business Management
4	Aman Verma	aman@gmail.com	92	Technology	Software Developer
5	Sneha Patel	sneha@gmail.com	81	Arts	Graphic Designer
6	Karan Singh	karan@gmail.com	69	Commerce	Accounting
7					
8					
9					



result

	A	B
1	User Name	Suggestion By AI
2	Tushar Thakur	Hello Tushar, It's great to connect with you at Career Compass! Your interest in Information Technology and Computer Applications, combined with your goal of becoming a software developer, sets a clear and exciting path forward. You've already got a fantastic head start with foundational programming skills like C, C++, JS, and Python, along with prac --- ### Career Roadmap: Tushar Thakur - Aspiring Software Developer **Current Status:** 12th Grade Pass, Aspiring Software Developer #### Career Outlook: How does their chosen path look in 2026? The field of software development is booming in 2026, and its growth shows no signs of slowing down. There's a consistently high demand for skilled developers across various specializations, from web and mobile applications to artificial intelligence, data science, and cloud computing. India, in particular, is a global IT hub, offering immens For someone like you, who is just starting out after 12th grade, the outlook is exceptionally positive *if you commit to continuous learning and formal education*. Companies are always seeking fresh talent with strong problem-solving abilities and a solid technical foundation. Your existing skills give you a significant advantage, but the journe #### Action Plan: Three specific steps to take this month (April 2026) This month is crucial for setting the foundation for your higher education and practical skill development. 1. **Research and Apply for Undergraduate Programs:** Given your 12th qualification, your immediate priority should be to secure admission into a relevant undergraduate degree. Research and apply to Bachelor of Technology (B.Tech) or Bachelor of Engineering (BE) programs in Computer Science, Information Technology, or Software 2. **Start a Personal Project Portfolio:** Leverage your existing skills (C, C++, JS, Python) by building small, tangible projects. For example: * Create a simple website using HTML, CSS, and JavaScript. * Develop a Python script for a practical task (e.g., file organizer, data parser). * Build a basic calculator or game using C++ or Python. * Experiment with n8n to automate a personal workflow. This demonstrates practical application of your knowledge and will be invaluable for future internships or job applications. Start by documenting these projects on platforms like GitHub. 3. **Dive Deeper into Foundational Concepts (Online Learning):** While you're waiting for college admissions, utilize free online resources. Pick one of your existing languages (e.g., Python or JavaScript) and follow a structured online course on platforms like freeCodeCamp, Coursera, or Udemy. Focus on understanding core programmin #### Skill Gap: Which technical skills should they learn next to increase efficiency? To evolve from a 12th-grade pass with basic programming knowledge to an efficient, job-ready software developer, here are key technical skills you should focus on acquiring next: 1. **Data Structures and Algorithms (DSA):** This is fundamental for efficient problem-solving and optimizing code performance. Mastering DSA is crucial for technical interviews and writing scalable software. 2. **Version Control Systems (Git & GitHub):** Essential for collaborative development, tracking changes, and managing codebases. It's an industry standard for all professional developers. 3. **Database Management (SQL/NoSQL):** Learn to interact with databases. SQL (e.g., MySQL, PostgreSQL) is foundational for relational databases, and understanding NoSQL databases (e.g., MongoDB) will also be beneficial. 4. **Web Development Frameworks:** Since you know JavaScript, dive into a popular frontend framework like **React, Angular, or Vue.js** to build dynamic user interfaces. For backend, explore **Node.js (with Express.js), Django (Python), or Flask (Python)** to build robust web applications. 5. **Object-Oriented Programming (OOP) Principles (Deep Dive):** While you know C++ and Python, a thorough understanding of OOP concepts (encapsulation, inheritance, polymorphism, abstraction) is vital for writing clean, modular, and maintainable code in professional environments. #### Job Market: 3 job titles they qualify for, including estimated entry-level salary in India. It's important to set realistic expectations: Direct entry into most professional "Software Developer" roles *immediately after 12th grade* is rare in India, as most companies require a bachelor's degree (B.Tech/BE/BCA/BSc CS) or a very robust, self-taught portfolio with substantial project experience. However, after pursuing higher educatio 1. **Junior Software Developer:** This is the most direct role after completing a relevant degree and showcasing strong foundational programming and problem-solving skills. * **Estimated Entry-Level Salary (India, 2026-2028):** ₹3,00,000 - ₹6,00,000 per annum 2. **Junior Web Developer (Frontend/Backend):** If you specialize in web development frameworks and build a strong portfolio of web projects. * **Estimated Entry-Level Salary (India, 2026-2028):** ₹2,50,000 - ₹5,50,000 per annum 3. **Junior Python Developer / Junior Automation Engineer:** Leveraging your Python and automation skills, focusing on backend development, scripting, or process automation. * **Estimated Entry-Level Salary (India, 2026-2028):** ₹2,50,000 - ₹5,00,000 per annum Remember, Tushar, consistency is key. Keep learning, keep building, and actively seek out opportunities to apply your knowledge. Your journey has just begun, and with your clear goals and existing skills, you're well-positioned for success! Best regards, Your Career Advisor at Career Compass

User's Database

	A	B	C	D	E	F	G	H	I	J
1	First Name	Middle Name	Last Name	Gender	Phone Number	Email ID	Current Qualification	Stream	Percentage / CGPA	Subjects of Interest
2	Tushar		Thakur	Male	9056866335	vanshth2005@gmail.com	12	12	78	The subject which interests me the most is information technology and computer applications
3	Tushar		Thakur	Male	9056866335	vanshth2005@gmail.com	12	12	78	The subject which interests me the most is information technology and computer applications
4	Aditi	T	Sharma	Female	606003635	sharmaaditi2006@gmail.c	12	12	88	Information technology and computer graphic design
5	Aditi	T	Sharma		60606060608	sharmaaditi2006@gmail.c	12	12	88	Information Technology and Computer Graphics
6	Aditi	T	Sharma		4435687102	sharma2006aditi@gmail.c	BCA	BCA	88	Information Technology and System Graphics
7	nhcn	dn	nfdd		dnngng	nngng@1f	ngng	ngng	35	ngn
8	abh	dndd	nfdd		dnngng	vanshth2005@gmail.com	ngng	ngng	35	ngn

Figure: 5.6.1 – Google Sheets Database

5.7 Recommendation Logic

The Career Compass system uses a rule-based recommendation mechanism to suggest suitable career options based on user input. The logic is implemented within Make.com using conditional operations and filters.

The recommendation system analyzes the academic performance and interests of the user to determine the most appropriate career path. This approach ensures that the suggestions are relevant, personalized, and easy to understand.

5.7.1 Example Logic

IF marks >= 85 AND interest = "Science"

THEN recommend "Engineering / Medical"

IF marks >= 75 AND interest = "Commerce"

THEN recommend "Business Management / Finance"

IF marks >= 65 AND interest = "Arts"

THEN recommend "Design / Humanities"

IF marks < 65

THEN recommend "Skill-based Courses / Vocational Training"

5.7.2 Processing Steps

The recommendation process follows a systematic sequence:

1. **Input Analysis:**
   The system reads the marks and interests provided by the user.
2. **Condition Evaluation:**
   Predefined rules are applied to evaluate the input data.
3. **Matching Criteria:**
   The system matches user data with suitable career categories.
4. **Filtering Options:**
   Irrelevant career options are filtered out based on conditions.
5. **Recommendation Generation:**
   The most suitable career option is selected and generated.
6. **Output Delivery:**
   The recommendation is sent back to the frontend and displayed to the user.

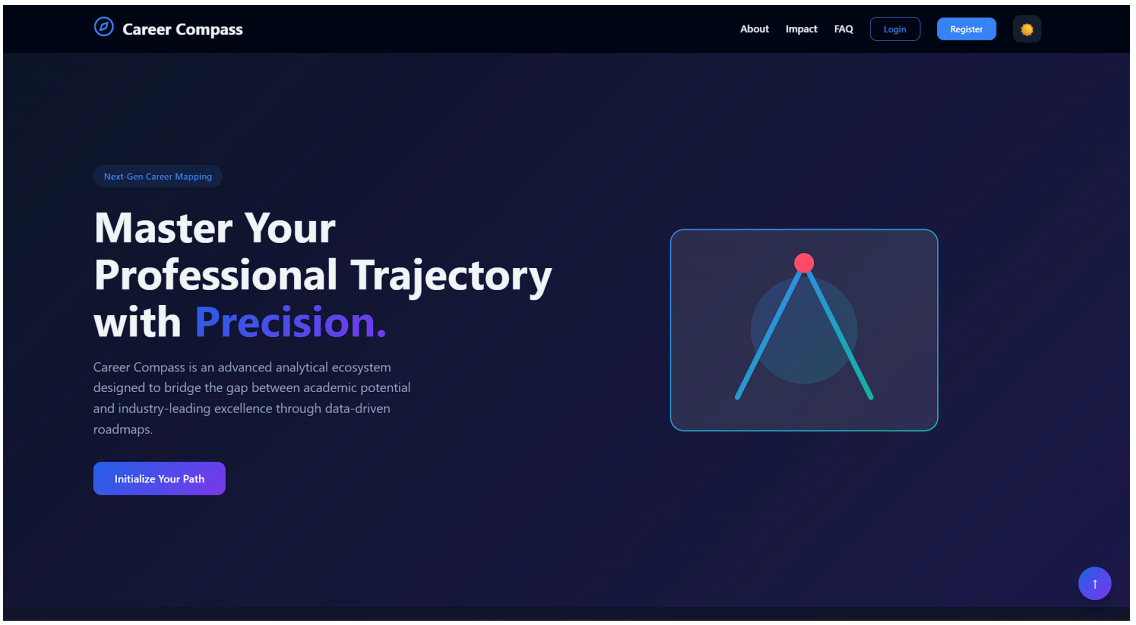
5.8 UI Design and Screenshots

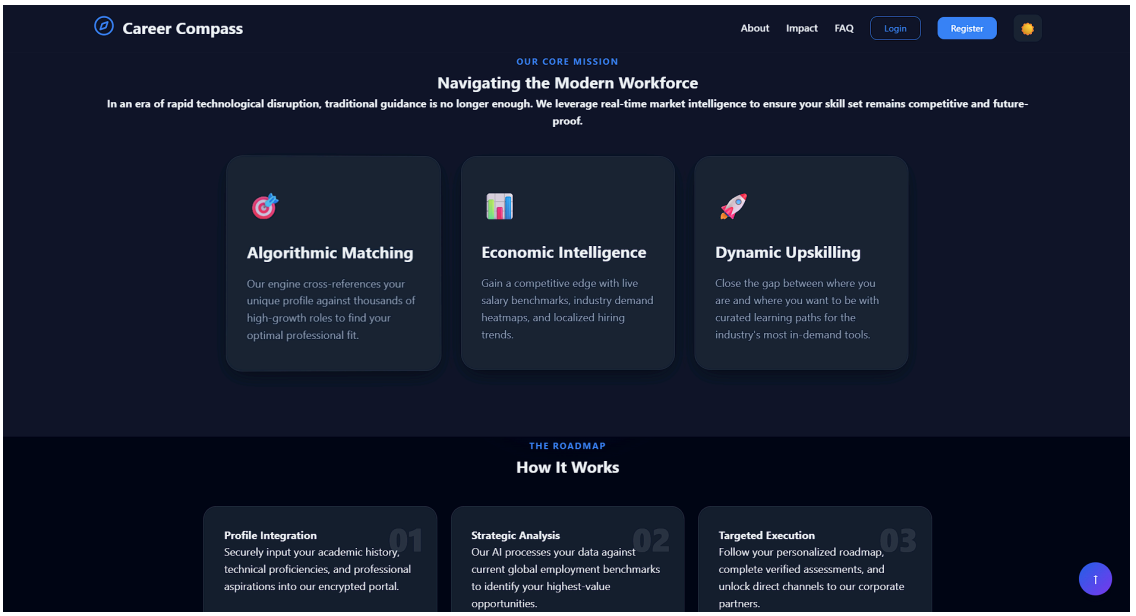
The user interface is designed to be simple, clean, and user-friendly to ensure ease of use for students.

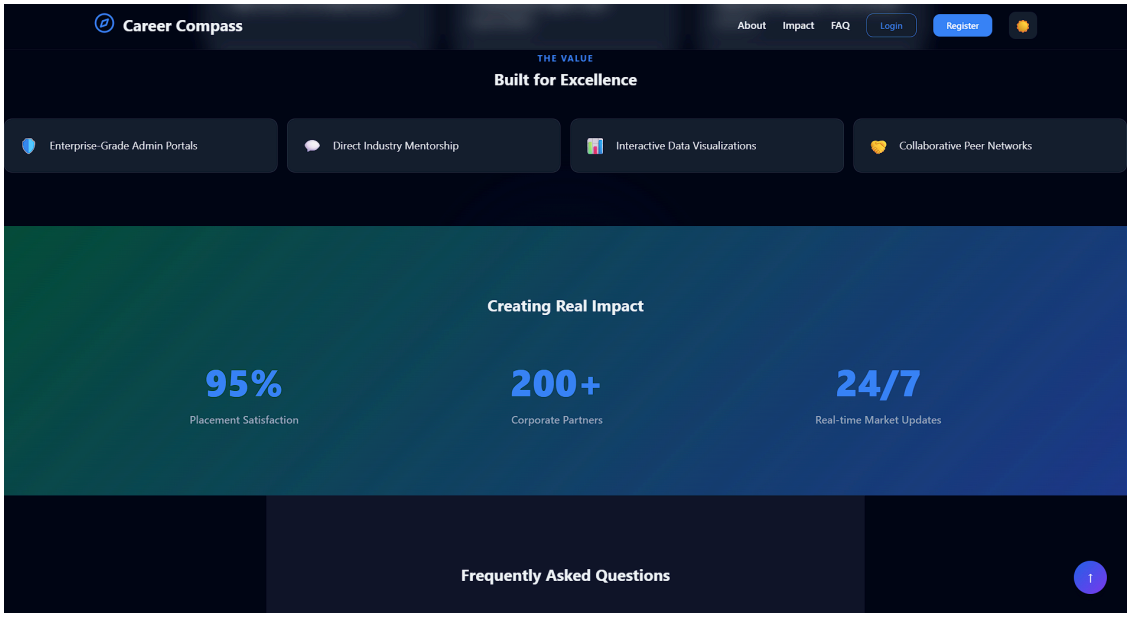
Screens Included:

- Main WebPage
- Login Page
- Registration Page

- Student Page
- Admin Page
- Student Input Form
- Result Page







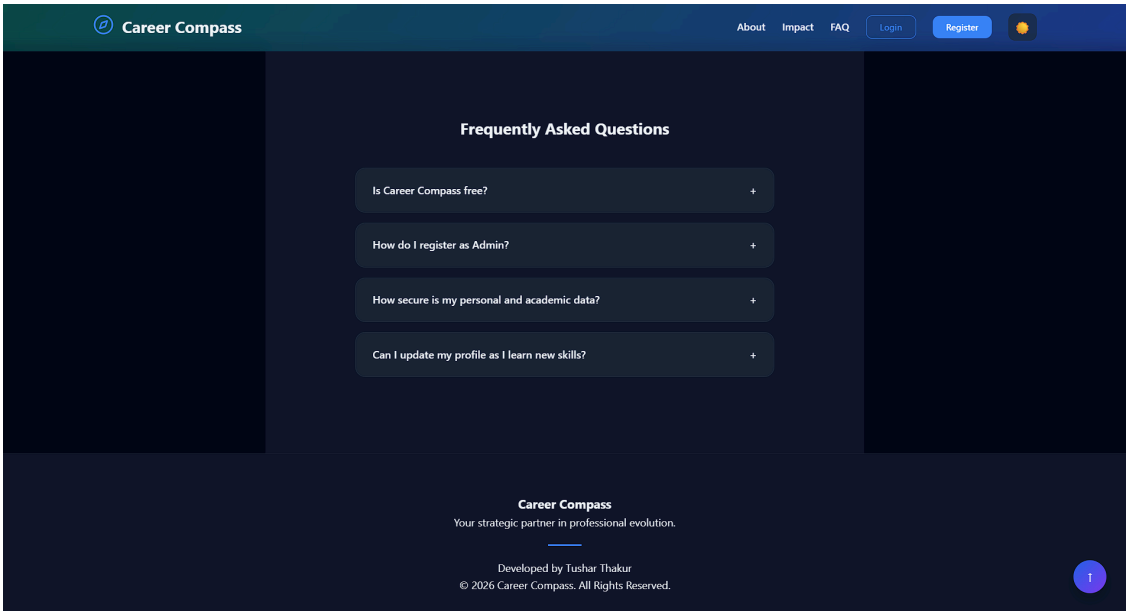


Figure: 5.8.1 – Main WebPage (Dark Mode)

Career Compass

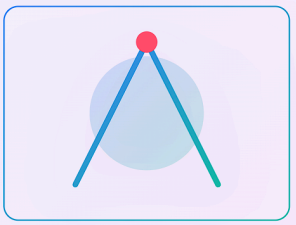
About Impact FAQ Login Register

Next-Gen Career Mapping

Master Your Professional Trajectory with Precision.

Career Compass is an advanced analytical ecosystem designed to bridge the gap between academic potential and industry-leading excellence through data-driven roadmaps.

Initialize Your Path



Career Compass

About Impact FAQ Login Register

OUR CORE MISSION

Navigating the Modern Workforce

In an era of rapid technological disruption, traditional guidance is no longer enough. We leverage real-time market intelligence to ensure your skill set remains competitive and future-proof.

Algorithmic Matching

Our engine cross-references your unique profile against thousands of high-growth roles to find your optimal professional fit.

Economic Intelligence

Gain a competitive edge with live salary benchmarks, industry demand heatmaps, and localized hiring trends.

Dynamic Upskilling

Close the gap between where you are and where you want to be with curated learning paths for the industry's most in-demand tools.

THE ROADMAP

How It Works

Profile Integration 01
Securely input your academic history, technical proficiencies, and professional aspirations into our encrypted portal.

Strategic Analysis 02
Our AI processes your data against current global employment benchmarks to identify your highest-value opportunities.

Targeted Execution 03
Follow your personalized roadmap, complete verified assessments, and unlock direct channels to our corporate partners.

Career Compass

About Impact FAQ Login Register

THE VALUE

Built for Excellence

Enterprise-Grade Admin Portals

Direct Industry Mentorship

Interactive Data Visualizations

Collaborative Peer Networks

Creating Real Impact

95%
Placement Satisfaction

200+
Corporate Partners

24/7
Real-time Market Updates

Frequently Asked Questions

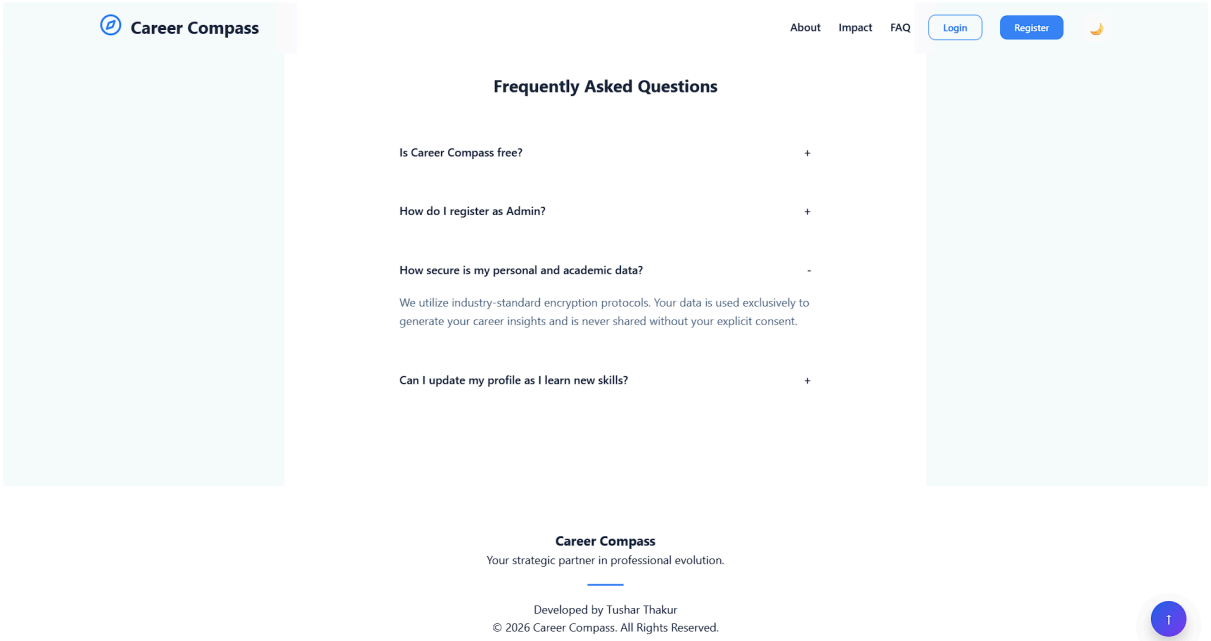


Figure: 5.8.2 – Main WebPage (Light Mode)

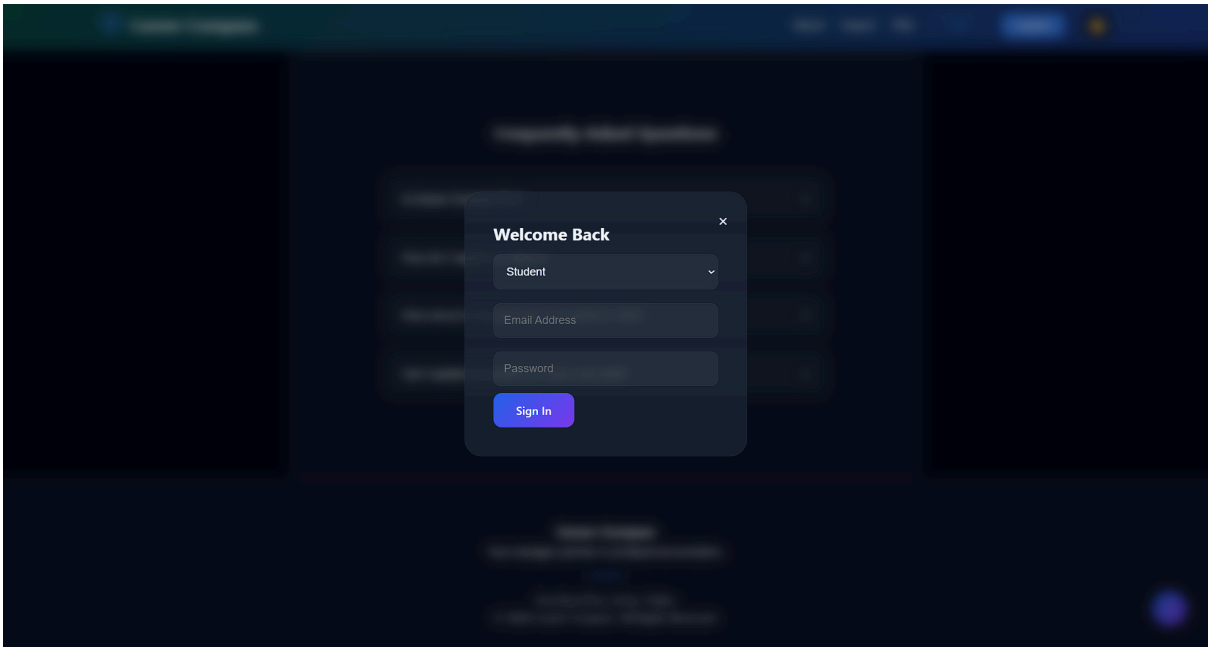


Figure: 5.8.3 – Login Page (Dark Mode)

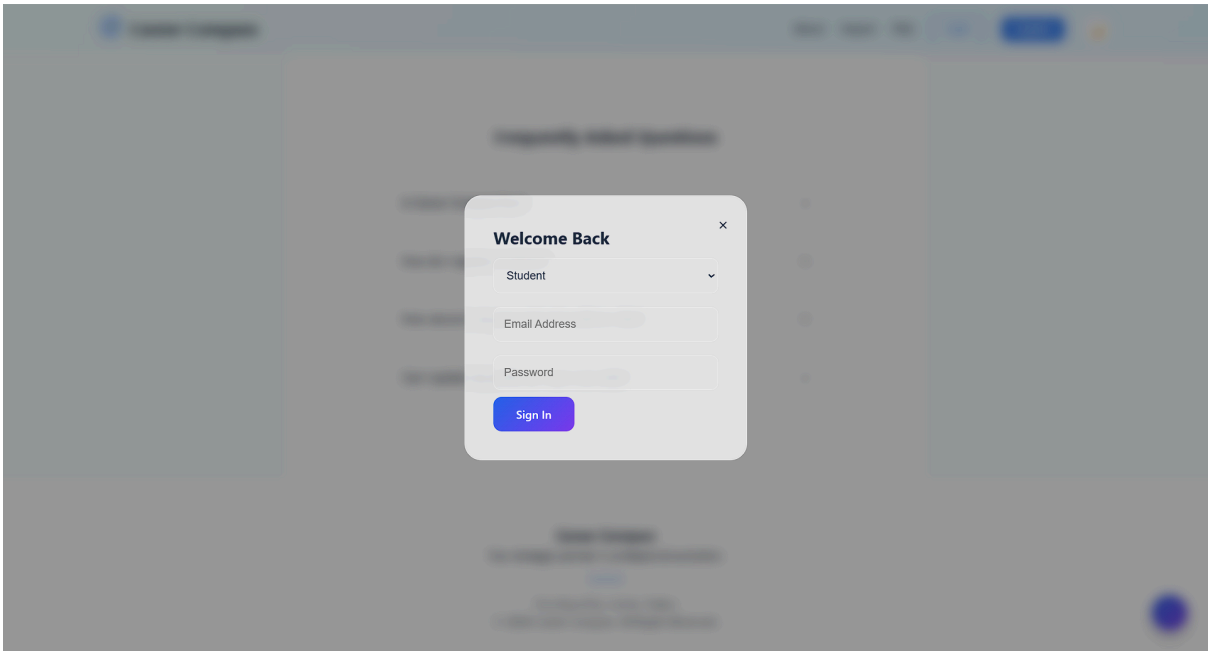


Figure: 5.8.4 – Login Page (Light Mode)

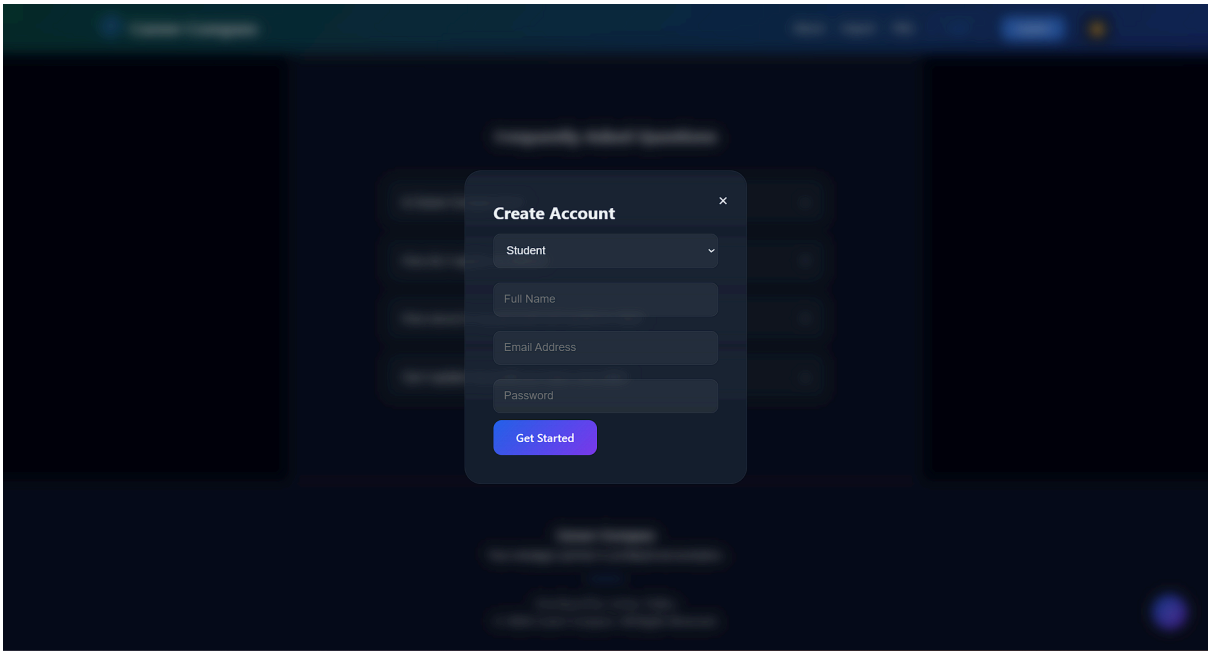


Figure: 5.8.5 – Registration Page (Dark Mode)

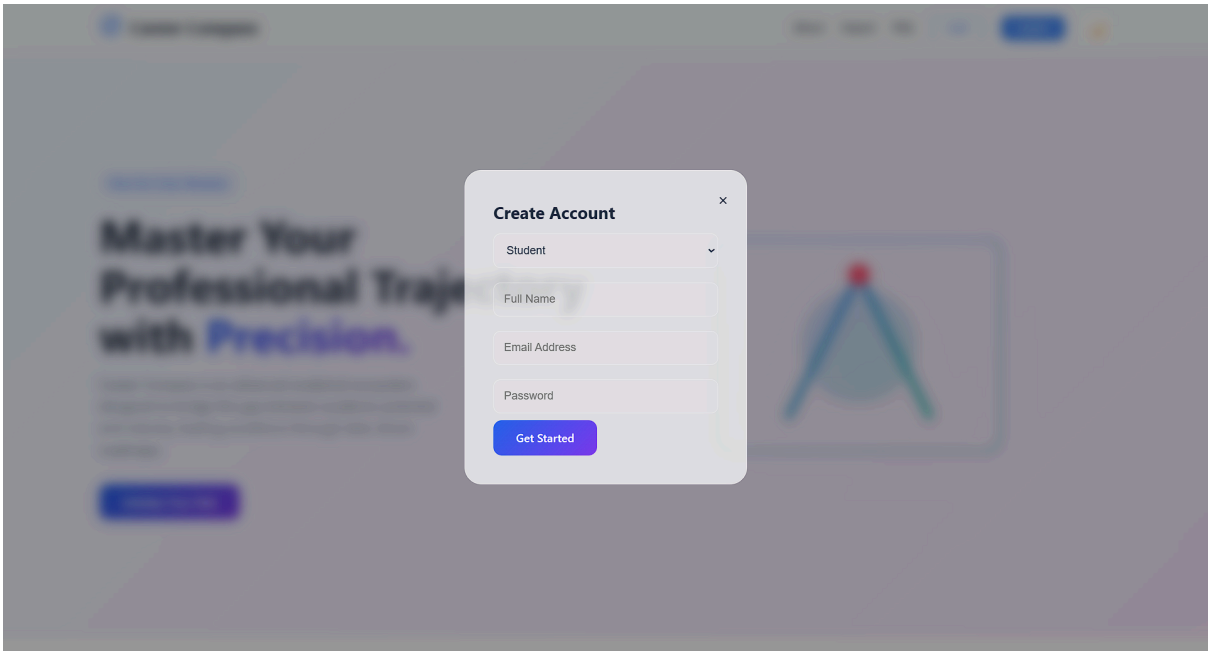


Figure: 5.8.6 – Registration Page (Light Mode)

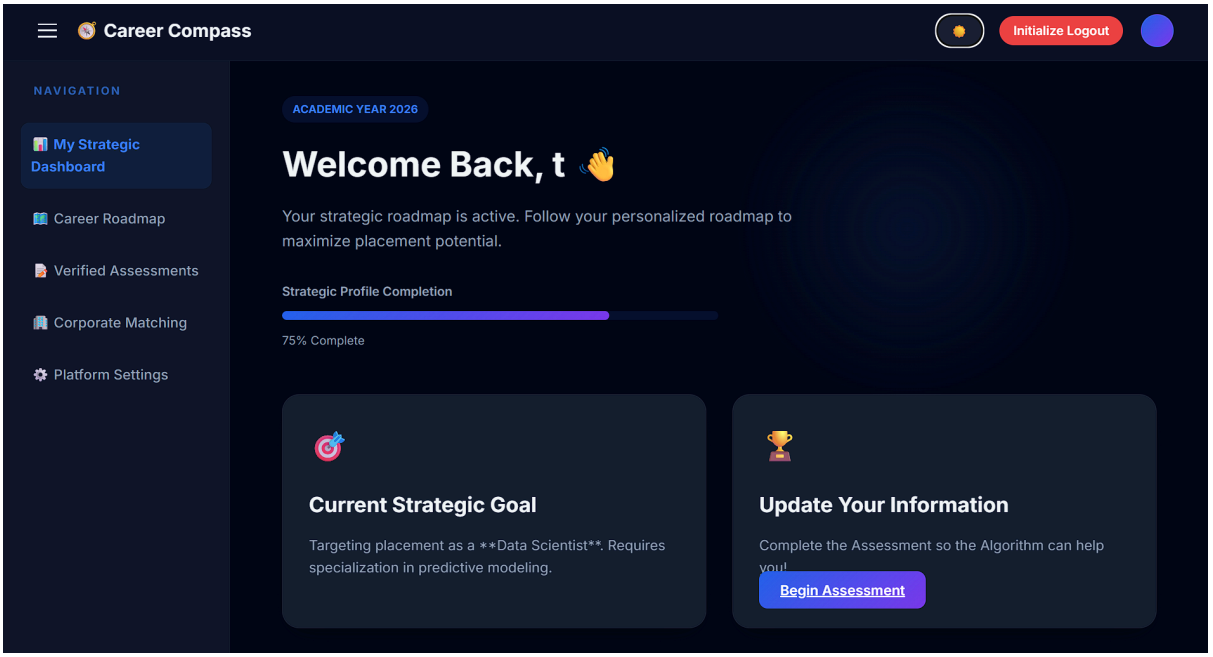


Figure: 5.8.7 – Student Page (Dark Mode)

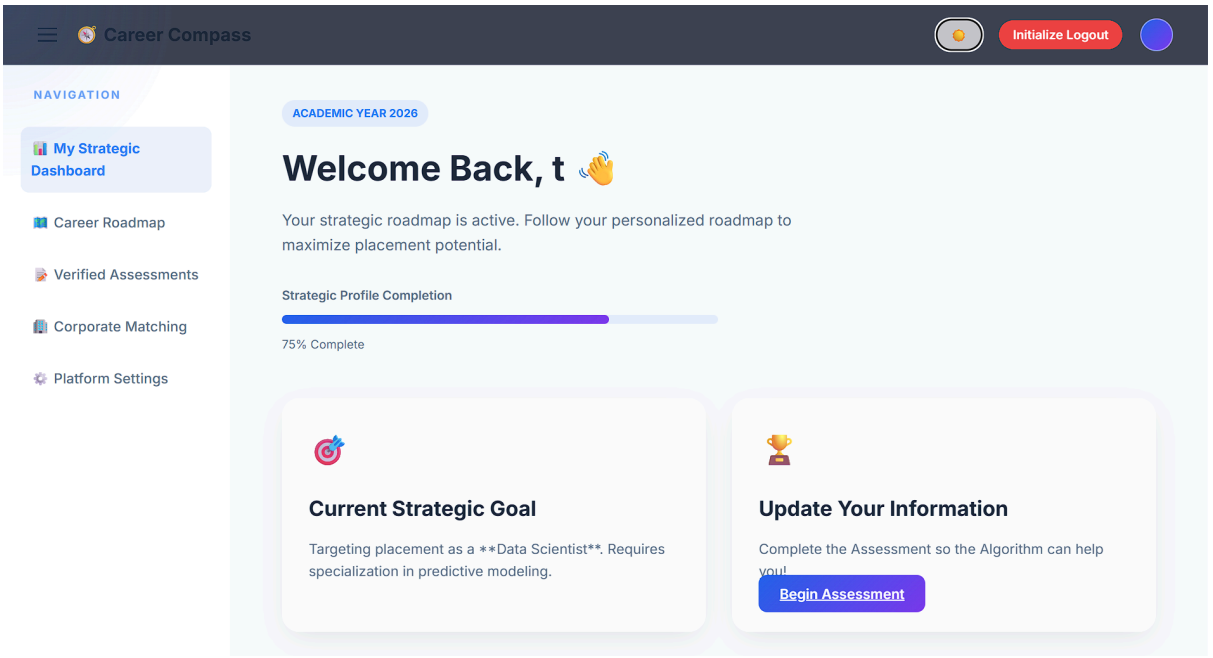


Figure: 5.8.8 – Student Page (Light Mode)

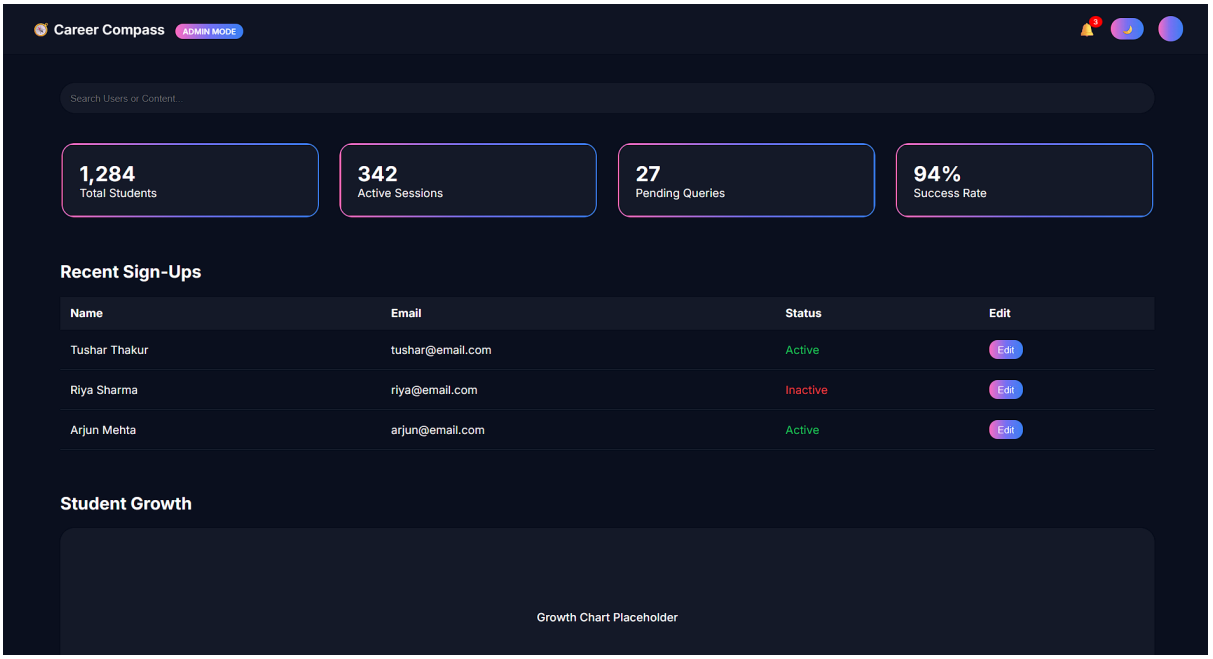


Figure: 5.8.9 – Admin Page (Dark Mode)

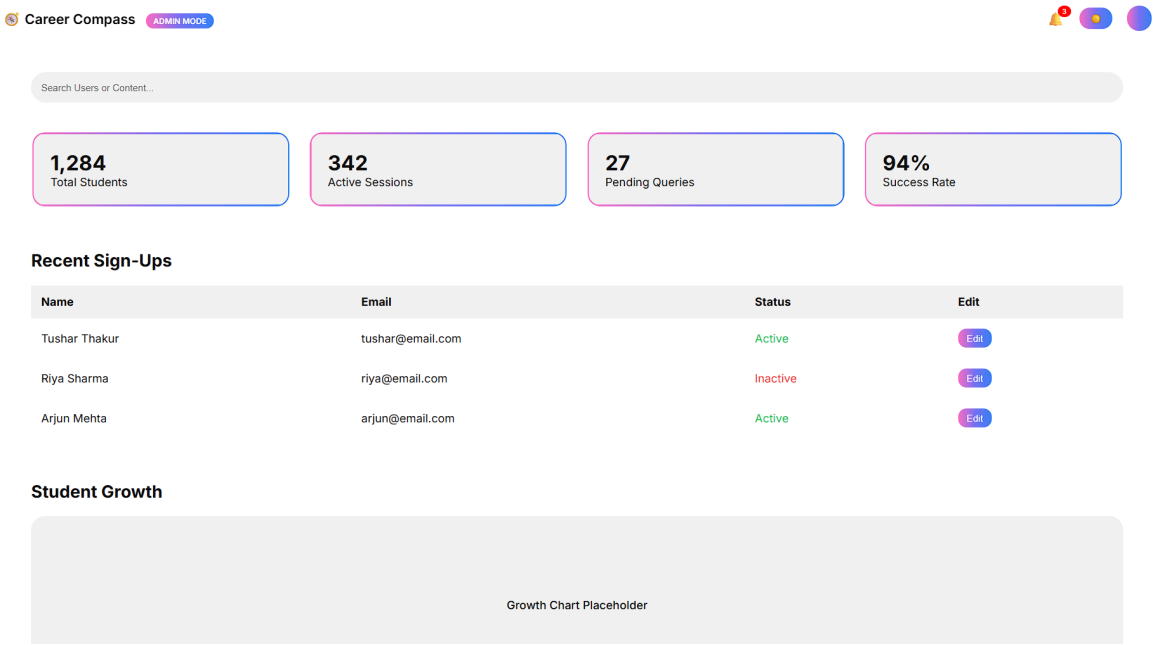


Figure: 5.8.10 – Admin Page (Light Mode)

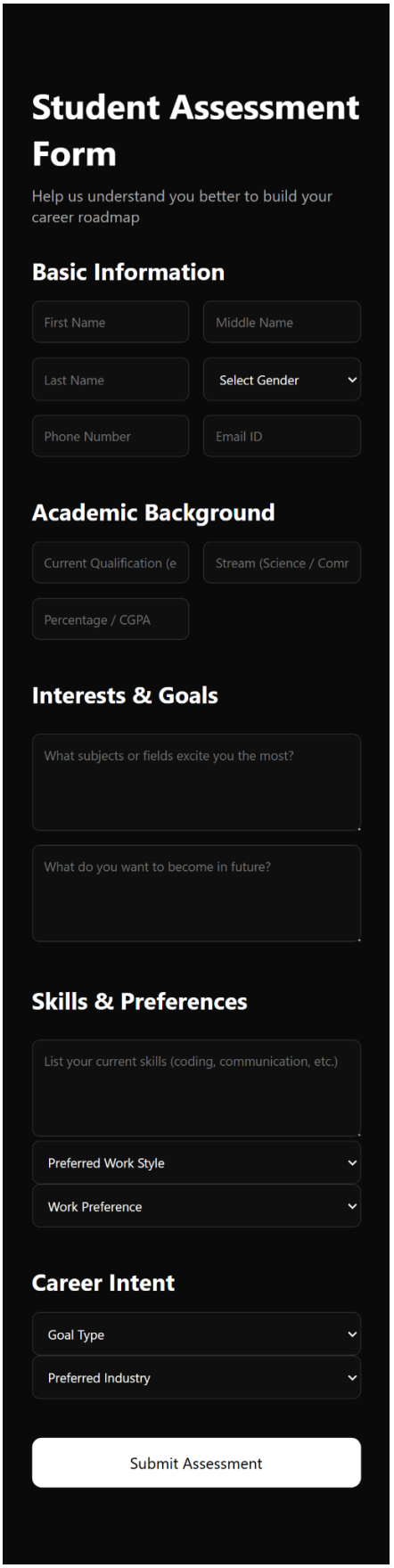


Figure: 5.8.11 – Input Form

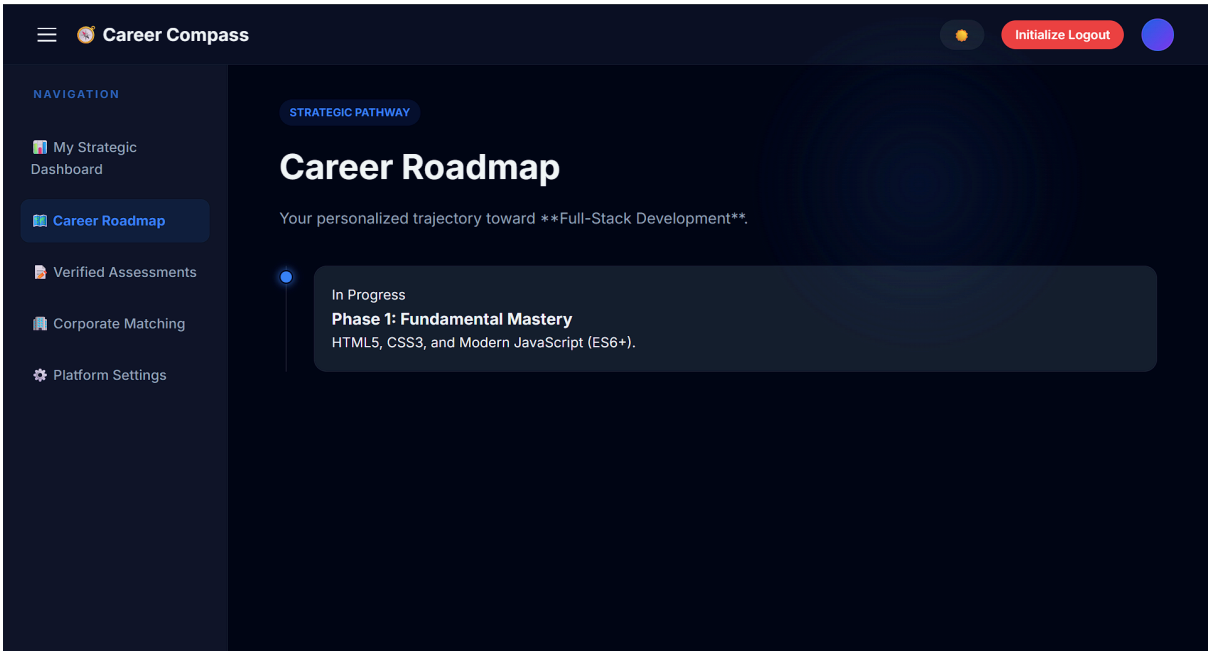


Figure: 5.8.12 – Result Displayed (Dark Mode)

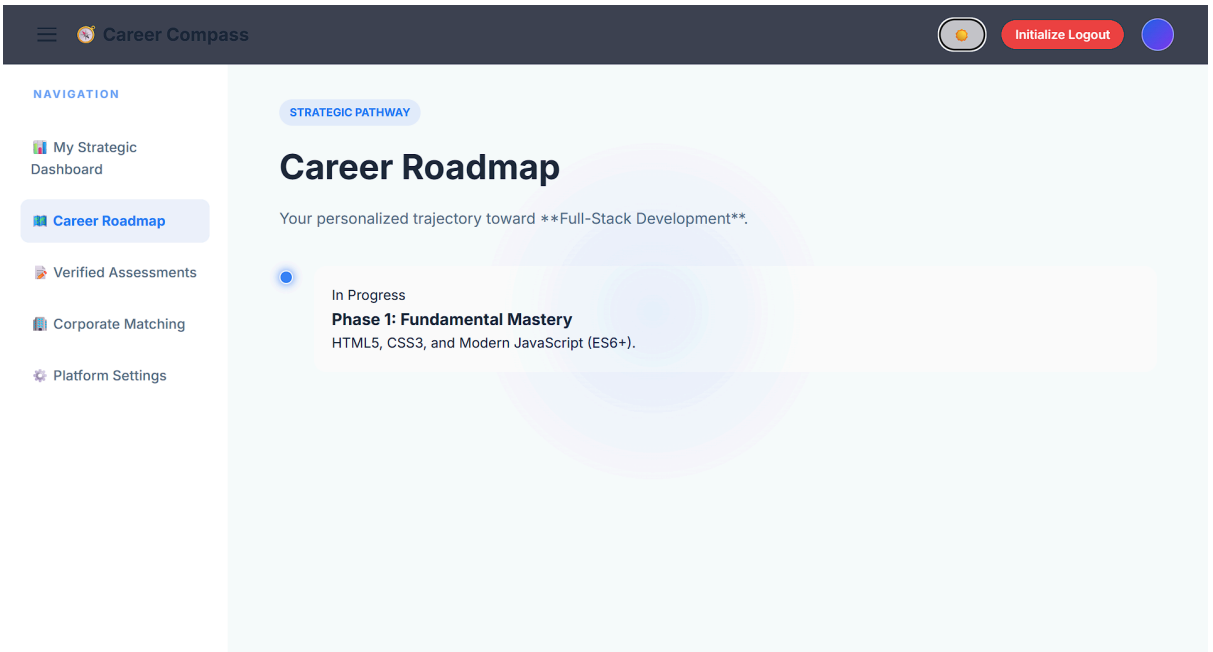


Figure: 5.8.13 – Result Displayed (Light Mode)

5.9 Testing of Modules

Testing is an essential phase to ensure that each component of the system functions correctly and efficiently. The Career Compass system undergoes module-wise testing to identify and resolve errors.

- **Frontend Testing:**
  Ensures that all forms, buttons, and user interface elements work correctly and provide a smooth user experience.
- **Data Submission Testing:**
  Verifies that user input is correctly captured and transmitted to Make.com without data loss.
- **Make.com Workflow Testing:**
  Checks whether the automation scenario executes properly, including webhook triggering, data processing, and integration with Google Sheets.
- **Database Testing (Google Sheets):**
  Ensures that data is accurately stored, updated, and retrieved from the sheet.
- **Output Verification:**
  Confirms that the generated career recommendations are accurate and displayed correctly to the user.

5.10 Challenges Faced

During the implementation of the system, several challenges were encountered:

- **Integration Complexity:**
  Establishing a smooth connection between the frontend and Make.com required proper configuration of webhooks and data formats.
- **Data Formatting Issues:**
  Ensuring that data sent from the frontend matched the expected format in Make.com and Google Sheets required careful mapping.
- **Handling Real-Time Updates:**
  Managing real-time data flow between components without delays was a challenge.

- **Using Google Sheets as a Database:**
  Since Google Sheets is not a traditional database, handling large data efficiently requires optimization.
- **Debugging Automation Workflows:**
  Identifying errors in Make.com scenarios required step-by-step testing and monitoring.

5.11 Advantages of Implementation

The implementation approach used in the Career Compass system offers several advantages:

- **No Need for Traditional Backend:**
  Eliminates the requirement for server-side programming and database management systems.
- **Ease of Integration:**
  Make.com provides seamless integration between frontend and Google Sheets.
- **Scalability:**
  The system can be easily expanded by modifying workflows and adding new rules.
- **Cost-Effective:**
  Reduces infrastructure and maintenance costs by using cloud-based tools.
- **User-Friendly Data Management:**
  Google Sheets allows easy viewing, editing, and management of stored data.
- **Rapid Development:**
  Automation tools enable faster implementation compared to traditional development methods.

Conclusion of Coding and Implementation

The Coding and Implementation phase successfully transforms the system design into a fully functional and interactive application. During this phase, all the planned modules such as user interface, data collection forms, automation workflows, and data storage mechanisms are developed and integrated to work seamlessly together.

By utilizing frontend technologies like HTML, CSS, and JavaScript, the system provides a user-friendly interface that allows users to easily input their academic details and preferences. The integration with Make.com enables efficient handling of application logic without the need for a traditional backend server. This automation platform processes user inputs, applies predefined rules, and generates appropriate career recommendations in real time.

Google Sheets serves as a cloud-based database, ensuring that all user data and recommendation results are stored securely and can be accessed or updated whenever required. This approach simplifies data management while maintaining reliability and scalability.

The modular structure of the system ensures that each component operates independently while contributing to the overall functionality. This makes the system easier to maintain, update, and expand in the future. Additionally, the use of automation reduces manual effort, minimizes errors, and improves overall system efficiency.

Overall, the implementation demonstrates how modern web technologies combined with cloud-based automation tools can be effectively used to build a scalable, cost-efficient, and user-centric application.

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6. Testing

Testing is a critical and comprehensive phase in the development of the Career Compass system. It ensures that the application functions correctly, efficiently, and reliably under various conditions and scenarios. The testing process plays a vital role in identifying errors, validating system performance, ensuring data accuracy, and confirming that all components of the system work together seamlessly.

In this project, testing was carried out in a structured and systematic manner to evaluate both individual modules and the complete integrated system. Special attention was given to validating the interaction between the frontend interface, Make.com automation workflows, and Google Sheets database.

The system was tested at multiple levels, including unit testing, integration testing, system testing, and user acceptance testing, to ensure robustness and reliability.

6.1 Testing Strategy

A well-defined testing strategy was adopted to ensure thorough evaluation of the system. The strategy focused on verifying functionality, performance, usability, and reliability of the Career Compass system.

6.1.1 Objectives of Testing

The primary objectives of testing the system are as follows:

- To ensure that all modules and components function as intended without errors
- To validate the accuracy and consistency of data processing
- To verify seamless integration between frontend, Make.com, and Google Sheets
- To identify, analyze, and eliminate bugs or inconsistencies
- To ensure that the system provides a smooth and user-friendly experience
- To evaluate system performance under different input conditions
- To confirm that the recommendation logic produces correct outputs

6.1.2 Types of Testing Used

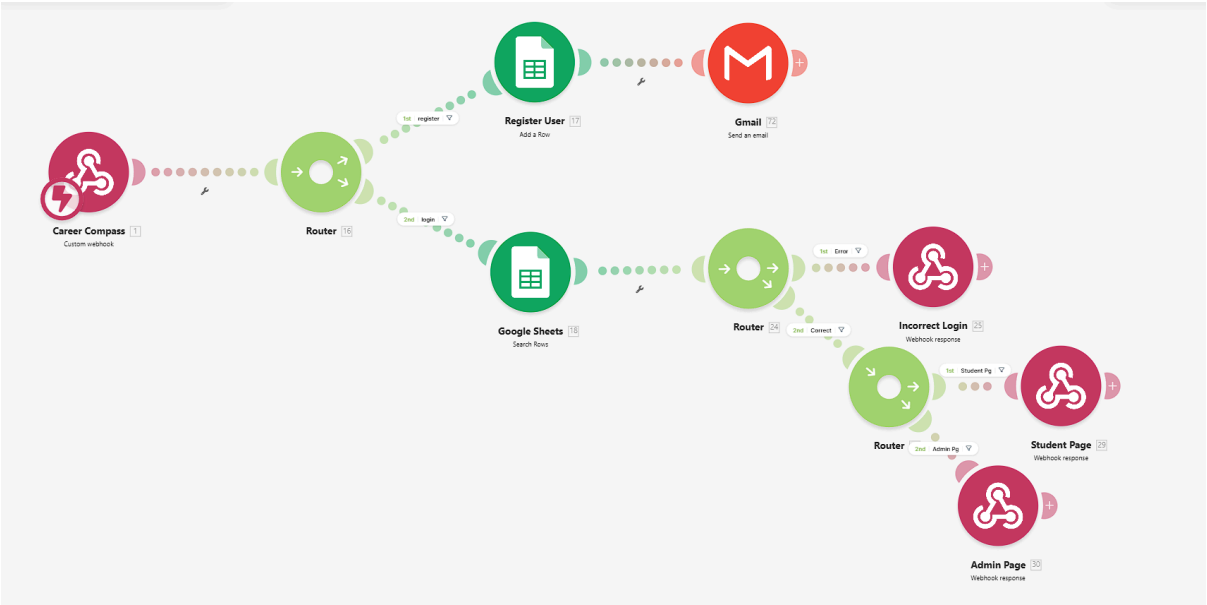
Multiple testing techniques were applied to ensure complete system validation:

- **Unit Testing:**
  Unit testing involves testing individual components or modules independently. Each part of the system, such as login forms, registration forms, input forms, and JavaScript functions, was tested separately to ensure correct functionality.
- **Integration Testing:**
  Integration testing verifies that different modules of the system interact correctly with each other. The following integrations were tested:
    - Frontend → Make.com (via Webhook)
    - Make.com → Google Sheets (data storage)
    - Google Sheets → Output display (data retrieval)
- This ensured smooth data flow across all components.
- **System Testing:**
  System testing evaluates the complete system as a whole. It checks whether the entire workflow—from user input to final output—is functioning correctly. This includes validating the recommendation logic and ensuring that the system behaves as expected under normal conditions.
- **User Acceptance Testing (UAT):**
  User Acceptance Testing was conducted with actual users (students) to evaluate usability, interface design, and overall satisfaction. Feedback was collected to improve system performance and user experience.

6.1.3 Testing Workflow

The testing workflow follows a structured sequence to ensure proper validation of each stage:

1. The user enters data into the frontend interface
2. Input validation is performed using JavaScript
3. Valid data is sent to Make.com via Webhook
4. Make.com processes the data using predefined logic
5. Processed data is stored in Google Sheets
6. Career recommendations are generated
7. Output is returned and displayed to the user



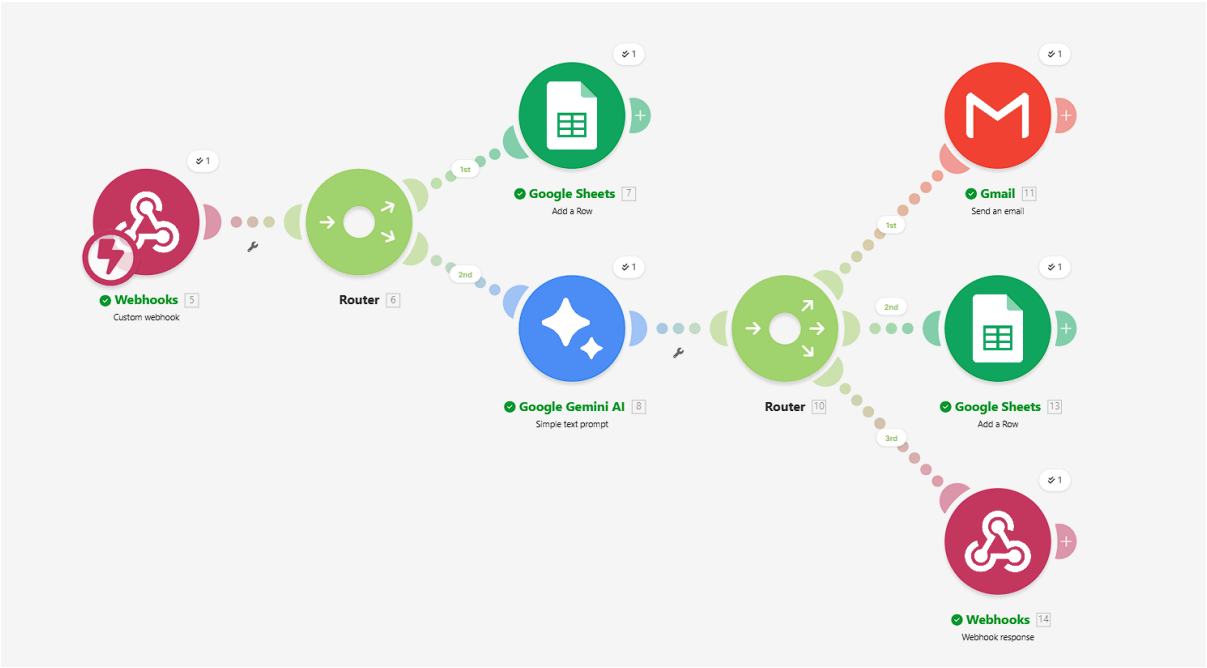


Figure: 6.1.1 – Testing Workflow

6.2 Test Cases

Test cases were designed to verify different functionalities and scenarios within the system. Each test case includes input conditions, expected output, actual output, and status.

Test Case ID	Test Description	Input	Expected Output	Actual Output	Status
TC01	Valid Login	Correct email & password	Login successful	Login successful	Pass
TC02	Invalid Login	Wrong credentials	Error message	Error displayed	Pass
TC03	Empty Form Submission	No input	Validation error	Error shown	Pass
TC04	Data Submission	Valid marks & interest	Data stored in Google Sheets	Stored correctly	Pass
TC05	Recommendation Logic	Marks = 90, Interest = Science	Engineering suggested	Engineering shown	Pass
TC06	API Failure	Disconnect internet	Error handled	Error message shown	Pass

Additional test cases were also performed to check boundary conditions, such as minimum and maximum marks, invalid data formats, and repeated submissions.

6.3 Test Results

The results obtained from testing indicate that the system performs efficiently and meets the intended objectives. All modules were tested successfully, and the system demonstrated reliable performance.

6.3.1 Functional Results

- All modules are functioning correctly without major errors
- Data is processed accurately and consistently
- Career recommendations are generated based on input conditions
- System workflow operates smoothly from input to output

6.3.2 Performance Results

- **Response Time:** The system responds quickly, typically within a few seconds
- **System Stability:** The system remains stable during repeated usage
- **Data Accuracy:** High level of accuracy in storing and retrieving data

6.3.3 Error Handling Results

- Invalid inputs are detected and handled properly
- Empty fields are restricted using validation techniques
- API failures and connectivity issues are managed effectively

6.3.4 User Feedback

Feedback collected from users indicates:

- The interface is simple and easy to use
- Navigation is clear and intuitive
- Career recommendations are helpful and relevant
- Overall user experience is satisfactory

6.4 Limitations Identified During Testing

Despite successful testing, some limitations were identified:

- The system depends on internet connectivity for operation
- Recommendation logic is rule-based and may not cover all scenarios
- Google Sheets performance may decrease with large datasets
- Limited scalability compared to advanced database systems

6.5 Improvements Made After Testing

Based on testing results and feedback, several improvements were implemented:

- Enhanced input validation to prevent incorrect data entry
- Optimized Make.com workflows for faster processing
- Improved user interface for better usability
- Refined recommendation logic for more accurate results

Conclusion of Testing Phase

The testing phase confirms that the Career Compass system operates effectively, efficiently, and reliably. All modules were validated successfully, and the system demonstrated accurate performance in generating career recommendations. The identified limitations are minimal and can be addressed in future enhancements. Overall, the system is robust, user-friendly, and ready for deployment.

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7. Implementation & Evaluation

This phase focuses on the practical deployment of the Career Compass system and evaluates its real-world performance. The implementation ensures that the designed system is successfully executed, while evaluation determines its efficiency, usability, and effectiveness.

7.1 System Implementation

The Career Compass system was implemented using a combination of frontend technologies and automation tools. The integration of the user interface with Make.com and Google Sheets enabled seamless data processing and storage.

7.1.1 Implementation Environment

The system was developed and tested in the following environment:

- **Operating System:** Windows 10/11
- **Frontend:** HTML, CSS, JavaScript
- **Automation Platform:** Make.com
- **Database:** Google Sheets
- **Browser:** Google Chrome

7.1.2 Deployment Architecture

The deployment architecture defines how different components interact in a real-world environment.

The system follows a cloud-based lightweight architecture:

1. User accesses the web interface
2. Inputs data through forms
3. Data is sent to Make.com via API/webhooks
4. Make.com processes and routes the data
5. Data is stored in Google Sheets
6. Results are generated and displayed

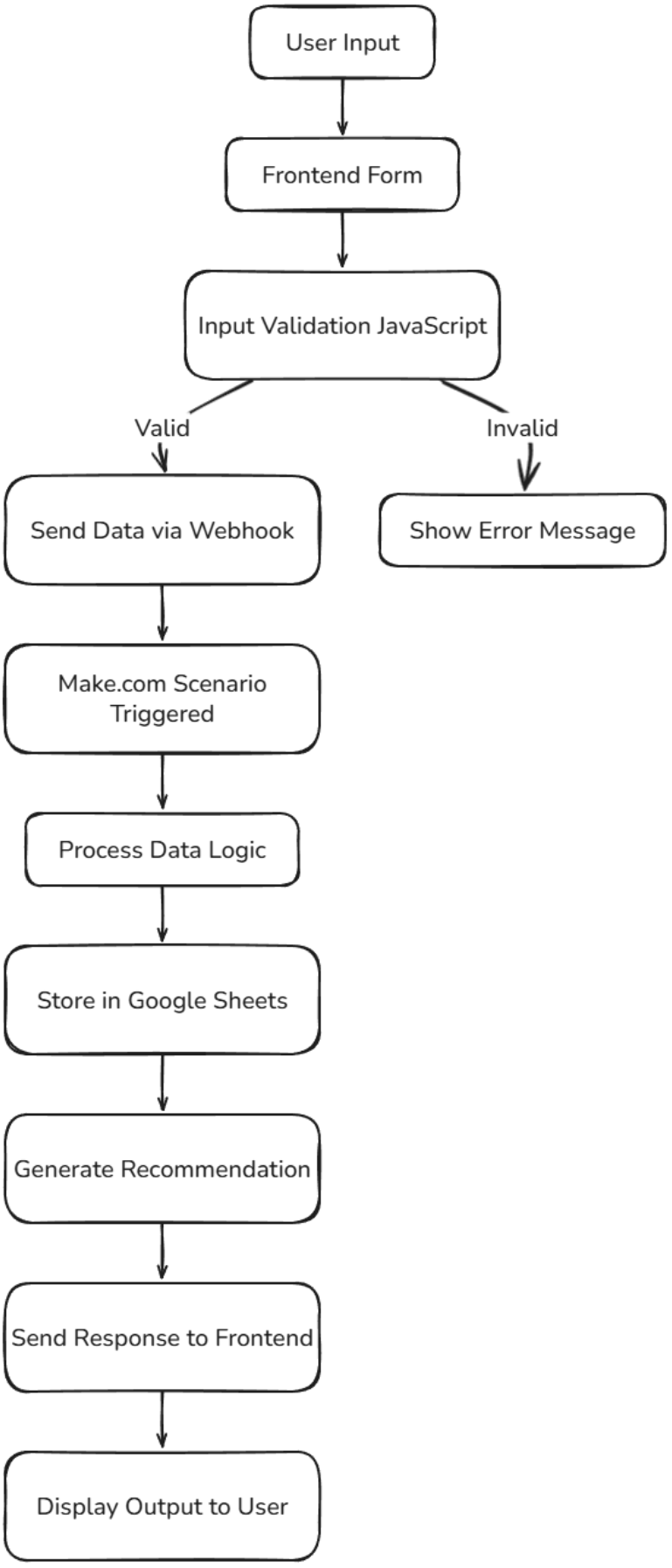


Figure: 7.1.1 – Deployment Architecture

7.1.3 Implementation Steps

The system was implemented step-by-step as follows:

1. **Frontend Development**
    - Designed responsive UI using HTML and CSS
    - Added validation using JavaScript
2. **Form Integration**
    - Created input forms for user data collection
    - Connected forms to Make.com webhooks
3. **Automation Setup (Make.com)**
    - Configured scenario workflows
    - Defined triggers and actions
    - Processed user input data
4. **Database Setup (Google Sheets)**
    - Created structured sheets
    - Defined columns for storing user data
    - Integrated with Make.com
5. **Testing & Debugging**
    - Verified data flow
    - Fixed errors in automation
    - Improved performance

7.1.4 Implementation Workflow

The workflow ensures proper execution of system operations:

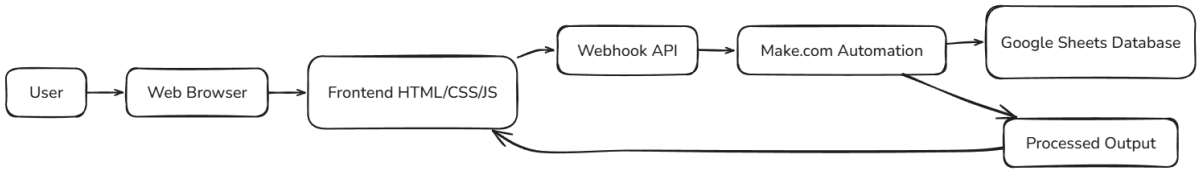


Figure: 7.1.2 – Implementation Workflow

7.2 System Evaluation

System evaluation measures how well the system performs in real-world conditions.

7.2.1 Evaluation Criteria

The system was evaluated based on:

- Functionality
- Performance
- Usability
- Reliability
- Scalability

7.2.2 Functional Evaluation

- All modules operate correctly
- Input validation works effectively
- Career recommendations are generated accurately

7.2.3 Performance Evaluation

- Fast response time
- Efficient data processing
- Smooth workflow execution

7.2.4 Usability Evaluation

- Simple and intuitive interface
- Easy navigation
- User-friendly design

7.2.5 Reliability Evaluation

- Stable system performance
- Consistent output generation
- Minimal system failures

7.2.6 Scalability Evaluation

- Can handle increasing users
- Google Sheets supports moderate data scaling
- Make.com workflows can be extended

7.3 Real-Time Implementation Results

The system was tested in real-time scenarios:

- Students successfully entered their data
- Data was processed instantly
- Career recommendations were displayed correctly

7.4 Advantages of the Implemented System

- Automated workflow (no manual intervention)
- Easy data storage and retrieval
- Cost-effective solution
- Quick deployment

7.5 Limitations of Implementation

- Dependent on internet connectivity
- Limited advanced analytics
- Google Sheets scalability constraints

7.6 Maintenance

System maintenance ensures smooth functioning over time.

7.6.1 Types of Maintenance

- **Corrective Maintenance:** Fixing errors
- **Adaptive Maintenance:** Updating system for new needs
- **Preventive Maintenance:** Improving performance

7.6.2 Maintenance Activities

- Updating Make.com workflows
- Cleaning Google Sheets data
- Enhancing UI features
- Monitoring system performance

Conclusion of Implementation & Evaluation

The implementation of the Career Compass system was successfully carried out using modern web technologies and automation tools. The system performs efficiently and meets its objectives of providing accurate career guidance. Evaluation results confirm that the system is reliable, user-friendly, and scalable for future enhancements.

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8. Conclusion

The Career Compass system was successfully designed, developed, and implemented to provide intelligent and user-friendly career guidance for students. The primary objective of the project was to create a platform that assists users in making informed career decisions based on their academic performance and personal interests.

The system integrates modern web technologies with automation tools such as Make.com and cloud-based storage using Google Sheets. This combination enabled the development of a lightweight, cost-effective, and efficient solution without the need for complex backend infrastructure.

Throughout the development process, all phases of the Software Development Life Cycle (SDLC) were followed systematically, including requirement analysis, system design, implementation, testing, and evaluation. Each phase contributed significantly to ensuring the quality and reliability of the final system.

The frontend interface, developed using HTML, CSS, and JavaScript, provides a simple and intuitive user experience. Users can easily input their data, and the system processes this information in real-time using Make.com workflows. The processed data is stored securely in Google Sheets, and relevant career recommendations are generated and displayed instantly.

The testing phase confirmed that the system performs efficiently under various conditions. All modules were validated successfully, and the integration between different components was found to be smooth and reliable. The system demonstrated fast response time, accurate data handling, and consistent output generation.

One of the key strengths of the Career Compass system is its automation capability. By leveraging Make.com, the system eliminates manual intervention and ensures seamless data processing. Additionally, the use of Google Sheets as a database provides flexibility and ease of data management.

However, the system also has certain limitations. It relies on internet connectivity for operation and uses rule-based logic for generating recommendations, which may not cover all possible scenarios. Despite these limitations, the system fulfills its intended purpose effectively.

In conclusion, the Career Compass system is a practical and efficient solution for career guidance. It successfully demonstrates how modern tools and technologies can be integrated to build a functional application. The project not only meets its objectives but also provides a strong foundation for future enhancements, making it a valuable contribution in the field of career guidance systems.

9. Future Scope

The Career Compass system has been successfully developed and implemented to provide basic career guidance based on user inputs. However, with advancements in technology and increasing user expectations, there are numerous opportunities to enhance and expand the system in the future.

This section outlines potential improvements and extensions that can make the system more powerful, scalable, and intelligent.

9.1 Integration of Artificial Intelligence

In the current system, career recommendations are based on predefined rules. In the future, Artificial Intelligence (AI) and Machine Learning (ML) techniques can be integrated to provide more accurate and personalized recommendations.

- Use machine learning models to analyze user behavior and preferences
- Provide dynamic career suggestions based on large datasets
- Continuously improve recommendations through learning algorithms

This would significantly enhance the intelligence and adaptability of the system.

9.2 Advanced Data Analytics

The system can be enhanced by incorporating data analytics features:

- Analyze trends in student choices
- Generate reports and insights
- Provide statistical data on career paths

This will help users and administrators make better decisions based on data-driven insights.

9.3 Mobile Application Development

To improve accessibility and usability, a mobile application can be developed:

- Android and iOS applications
- User-friendly mobile interface
- Real-time notifications

This would allow users to access the system anytime and anywhere.

9.4 User Authentication and Security

Currently, the system has limited security features. Future improvements can include:

- Secure login and authentication system
- Data encryption
- Role-based access control (Admin/User)

This will ensure data privacy and system security.

9.5 Integration with External APIs

The system can be extended by integrating external APIs:

- Career databases
- Job market trends
- Educational course platforms

This will provide more comprehensive and up-to-date career guidance.

9.6 Chatbot Integration

A chatbot can be added to assist users:

- Answer user queries instantly

- Guide users through the system
- Provide interactive support

This will improve user engagement and experience.

9.7 Improved Recommendation System

The recommendation logic can be enhanced by:

- Including more parameters (skills, personality, interests)
- Providing multiple career options
- Ranking recommendations based on suitability

This will make the system more effective and user-centric.

9.8 Cloud-Based Database Upgrade

Instead of relying solely on Google Sheets, future versions can use:

- Cloud databases (e.g., Firebase, AWS, or SQL-based systems)
- Faster data processing
- Better scalability and performance

9.9 Multi-language Support

To make the system accessible to a wider audience:

- Add support for multiple languages
- Provide regional language options

This will increase usability across different regions.

9.10 Admin Dashboard

An advanced admin panel can be developed:

- Monitor user data
- Analyze system performance
- Manage recommendations

This will improve system management and control.

9.11 Integration with Educational Platforms

The system can be connected with:

- Online learning platforms
- Course recommendation systems
- Skill development resources

This will help users take actionable steps after receiving career guidance.

9.12 Integration with Power BI for Reporting

In the future, the system can be enhanced by integrating Microsoft Power BI to provide advanced data visualization and reporting features.

- Generate detailed student performance reports
- Provide graphical dashboards (charts, graphs, insights)
- Allow users to download reports in PDF or Excel format
- Enable data-driven decision making

This feature will help students better understand their strengths and career possibilities through visual analytics.

9.13 Integration with LinkedIn

The system can be integrated with LinkedIn to enhance career recommendations and professional connectivity.

- Suggest relevant LinkedIn profiles based on career interests
- Recommend industry professionals and mentors
- Provide insights into career paths and professional journeys
- Allow users to connect directly with professionals

This integration will bridge the gap between career guidance and real-world networking.

9.14 Job Recommendation System

A job recommendation module can be introduced to provide career opportunities along with guidance.

- Recommend jobs based on user skills and interests
- Display current job openings in relevant fields
- Provide job descriptions and required qualifications
- Suggest internships and entry-level opportunities

This will make the system more practical by helping users take the next step after choosing a career path.

9.15 Integration with Job Portals

The system can be connected with popular job portals to fetch real-time job data.

- Integration with platforms like Naukri, Indeed, or LinkedIn Jobs
- Display live job listings
- Filter jobs based on location, skills, and qualifications

This will ensure that users receive up-to-date career opportunities.

9.16 Personalized Career Roadmap

A personalized roadmap feature can be introduced to guide users step-by-step.

- Suggest courses and certifications
- Provide skill development plans
- Set short-term and long-term career goals
- Track user progress

This will transform the system from a recommendation tool into a complete career planning platform.

Conclusion of Future Scope

The Career Compass system has significant potential for growth and enhancement. By integrating advanced technologies such as artificial intelligence, mobile applications, and cloud computing, the system can evolve into a comprehensive career guidance platform. These improvements will not only increase the system's efficiency and scalability but also enhance user experience and satisfaction. With the addition of advanced features such as Power BI reporting, LinkedIn integration, and job recommendation systems, the Career Compass platform can evolve into a comprehensive ecosystem for career development. These enhancements will not only improve the accuracy of recommendations but also provide users with actionable insights, professional networking opportunities, and real-world job access.

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10. Annexure

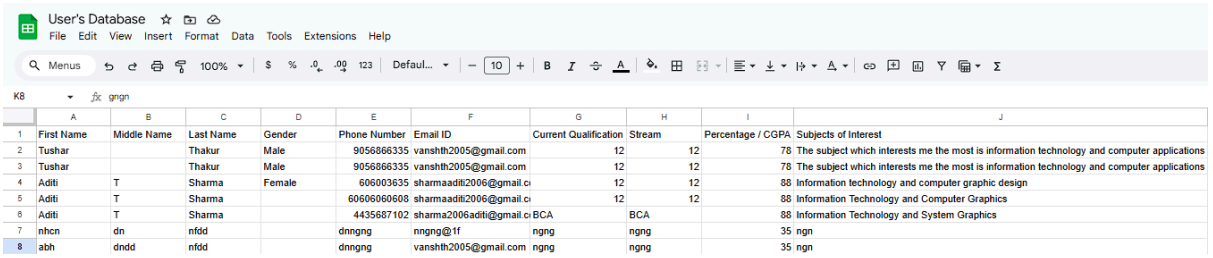
The Annexure section provides supplementary material that supports the development and implementation of the Career Compass System. This section includes structured data formats, sample configurations, extended code references, workflow logic, and supporting technical details. It is designed to enhance understanding of the system without relying on visual screenshots or images.

10.1 Sample User Data Structure (Google Sheets)

The system uses Google Sheets as a database, where user data is stored in a tabular format.

	A	B	C	D	E
1	**Name**	**Email**	**Marks**	**Interests**	**Recommended Career**
2	Rahul Sharma	rahul@gmail.com	88	Science	Engineering
3	Priya Mehta	priya@gmail.com	76	Commerce	Business Management
4	Aman Verma	aman@gmail.com	92	Technology	Software Developer
5	Sneha Patel	sneha@gmail.com	81	Arts	Graphic Designer
6	Karan Singh	karan@gmail.com	69	Commerce	Accounting
7					
8					
9					

10.2 Assessment Data Structure



	A	B	C	D	E	F	G	H	I	J
1	First Name	Middle Name	Last Name	Gender	Phone Number	Email ID	Current Qualification	Stream	Percentage / CGPA	Subjects of Interest
2	Tushar		Thakur	Male	9056866335	vanshth2005@gmail.com	12	12	78	The subject which interests me the most is information technology and computer applications
3	Tushar		Thakur	Male	9056866335	vanshth2005@gmail.com	12	12	78	The subject which interests me the most is information technology and computer applications
4	Aditi	T	Sharma	Female	606003635	sharmaaditi2006@gmail.c	12	12	88	Information technology and computer graphic design
5	Aditi	T	Sharma		60606060608	sharmaaditi2006@gmail.c	12	12	88	Information Technology and Computer Graphics
6	Aditi	T	Sharma		4435687102	sharma2006aditi@gmail.c	BCA	BCA	88	Information Technology and System Graphics
7	nhcn	dn	nfdd		dnngng	nngng@1f	ngng	ngng	35	ngn
8	abh	dndd	nfdd		dnngng	vanshth2005@gmail.com	ngng	ngng	35	ngn

10.3 Make.com Workflow Logic

The automation backbone of the system is built using Make.com scenarios.

Workflow Steps:

1. Webhook Trigger

- Receives data from frontend forms (login, register, assessment)

2. Router

- Divides flow into:
  - Registration
  - Login validation
  - Assessment processing

3. Google Sheets Modules

- Add new user (registration)
- Search user (login validation)
- Store assessment results

4. Conditional Filters

- Validate credentials
- Ensure no duplicate users

- Route correct responses

5. Response Handling

- Sends success/failure response back to frontend

10.4 Webhook Data Format (JSON)

User Registration Payload:

```
{
  "name": "Tushar Thakur",
  "email": "tushar@email.com",
  "password": "123456",
  "stream": "Science",
  "marks": "88",
  "interests": "Technology"
}
```

Login Payload:

```
{
  "email": "tushar@email.com",
  "password": "123456"
```

}

Assessment Submission Payload:

{

"email": "tushar@email.com",

"aptitude_score": 75,

"interest_score": 82,

"recommendation": "Software Developer"

}

10.5 Validation Rules Used

To maintain data integrity, the following validation rules are applied:

- Email must follow standard format (example@domain.com)

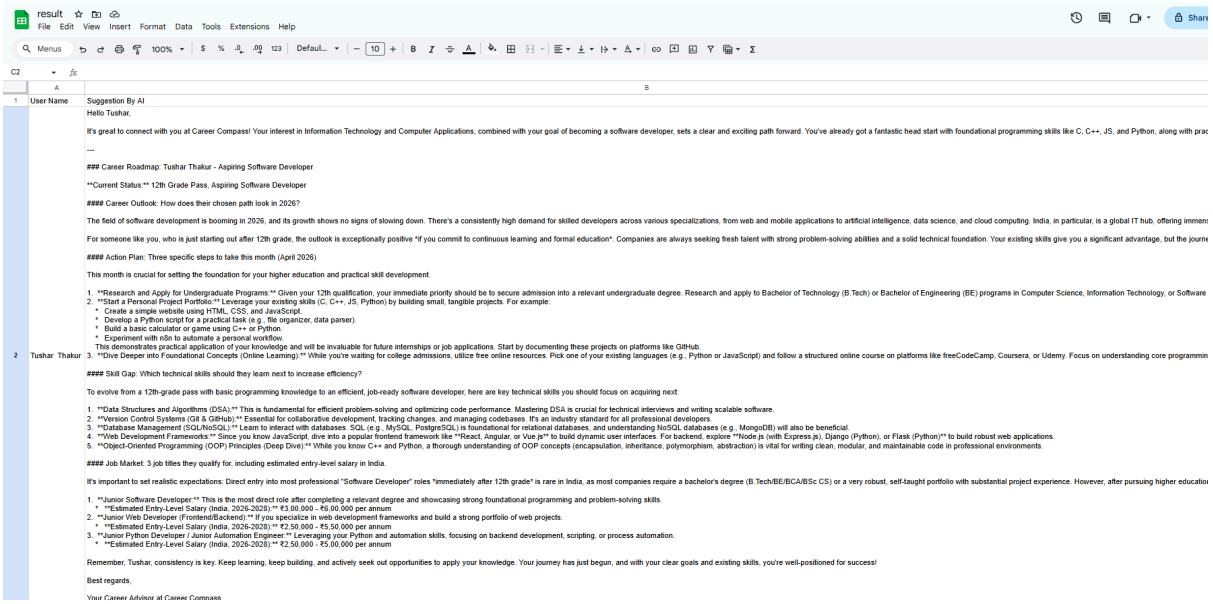
- Password must be minimum 6 characters

- Marks must be numeric (0–100)

- All fields are mandatory during registration

- Duplicate email entries are restricted

10.6 Error Handling Cases



result

File Edit View Insert Format Data Tools Extensions Help

	A	B
1	User Name	Suggestion By AI
2	Tushar Thakur	Hello Tushar, It's great to connect with you at Career Compass! Your interest in Information Technology and Computer Applications, combined with your goal of becoming a software developer, sets a clear and exciting path forward. You've already got a fantastic head start with foundational programming skills like C, C++, JS, and Python, along with prac --- ### Career Roadmap: Tushar Thakur - Aspiring Software Developer **Current Status:** 12th Grade Pass, Aspiring Software Developer #### Career Outlook: How does their chosen path look in 2026? The field of software development is booming in 2026, and its growth shows no signs of slowing down. There's a consistently high demand for skilled developers across various specializations, from web and mobile applications to artificial intelligence, data science, and cloud computing. India, in particular, is a global IT hub, offering immens For someone like you, who is just starting out after 12th grade, the outlook is exceptionally positive *if you commit to continuous learning and formal education*. Companies are always seeking fresh talent with strong problem-solving abilities and a solid technical foundation. Your existing skills give you a significant advantage, but the journe #### Action Plan: Three specific steps to take this month (April 2026) This month is crucial for setting the foundation for your higher education and practical skill development. 1. **Research and Apply for Undergraduate Programs:** Given your 12th qualification, your immediate priority should be to secure admission into a relevant undergraduate degree. Research and apply to Bachelor of Technology (B.Tech) or Bachelor of Engineering (BE) programs in Computer Science, Information Technology, or Software 2. **Start a Personal Project Portfolio:** Leverage your existing skills (C, C++, JS, Python) by building small, tangible projects. For example: * Create a simple website using HTML, CSS, and JavaScript. * Develop a Python script for a practical task (e.g., file organizer, data parser). * Build a basic calculator or game using C++ or Python. * Experiment with n8n to automate a personal workflow. This demonstrates practical application of your knowledge and will be invaluable for future internships or job applications. Start by documenting these projects on platforms like GitHub. 3. **Dive Deeper into Foundational Concepts (Online Learning):** While you're waiting for college admissions, utilize free online resources. Pick one of your existing languages (e.g., Python or JavaScript) and follow a structured online course on platforms like freeCodeCamp, Coursera, or Udemy. Focus on understanding core programmin #### Skill Gap: Which technical skills should they learn next to increase efficiency? To evolve from a 12th-grade pass with basic programming knowledge to an efficient, job-ready software developer, here are key technical skills you should focus on acquiring next: 1. **Data Structures and Algorithms (DSA):** This is fundamental for efficient problem-solving and optimizing code performance. Mastering DSA is crucial for technical interviews and writing scalable software. 2. **Version Control Systems (Git & GitHub):** Essential for collaborative development, tracking changes, and managing codebases. It's an industry standard for all professional developers. 3. **Database Management (SQL/NoSQL):** Learn to interact with databases. SQL (e.g., MySQL, PostgreSQL) is foundational for relational databases, and understanding NoSQL databases (e.g., MongoDB) will also be beneficial. 4. **Web Development Frameworks:** Since you know JavaScript, dive into a popular frontend framework like **React, Angular, or Vue.js** to build dynamic user interfaces. For backend, explore **Node.js (with Express.js), Django (Python), or Flask (Python)** to build robust web applications. 5. **Object-Oriented Programming (OOP) Principles (Deep Dive):** While you know C++ and Python, a thorough understanding of OOP concepts (encapsulation, inheritance, polymorphism, abstraction) is vital for writing clean, modular, and maintainable code in professional environments. #### Job Market: 3 job titles they qualify for, including estimated entry-level salary in India. It's important to set realistic expectations: Direct entry into most professional "Software Developer" roles *immediately after 12th grade* is rare in India, as most companies require a bachelor's degree (B.Tech/BE/BCA/BSc CS) or a very robust, self-taught portfolio with substantial project experience. However, after pursuing higher educatio 1. **Junior Software Developer:** This is the most direct role after completing a relevant degree and showcasing strong foundational programming and problem-solving skills. * **Estimated Entry-Level Salary (India, 2026-2028):** ₹3,00,000 - ₹6,00,000 per annum 2. **Junior Web Developer (Frontend/Backend):** If you specialize in web development frameworks and build a strong portfolio of web projects. * **Estimated Entry-Level Salary (India, 2026-2028):** ₹2,50,000 - ₹5,50,000 per annum 3. **Junior Python Developer / Junior Automation Engineer:** Leveraging your Python and automation skills, focusing on backend development, scripting, or process automation. * **Estimated Entry-Level Salary (India, 2026-2028):** ₹2,50,000 - ₹5,00,000 per annum Remember, Tushar, consistency is key. Keep learning, keep building, and actively seek out opportunities to apply your knowledge. Your journey has just begun, and with your clear goals and existing skills, you're well-positioned for success! Best regards, Your Career Advisor at Career Compass

10.7 Security Considerations (Basic Level)

Although full security implementation is planned in future phases, the current system includes:

- Basic input validation

- Controlled access via login

- Limited exposure of data fields

- Webhook endpoint privacy (hidden URL)

10.8 Deployment Details

- Frontend hosted locally or via static hosting platforms

- Google Sheets used as cloud database

- Make.com handles backend automation

- System accessible via browser interface

10.9 Limitations of Current System

- No encryption for passwords

- Limited scalability due to Google Sheets usage

- Basic recommendation logic (rule-based)

- No real-time analytics dashboard yet

10.10 Supporting Tools Used

- HTML, CSS, JavaScript – Frontend Development

- Make.com – Workflow Automation

- Google Sheets – Data Storage

- Webhooks – Data Communication

Conclusion of Annexure

This annexure provides a detailed technical foundation of the Career Compass system. It outlines the internal structure, data handling mechanisms, and automation workflows that power the application. These supporting details ensure clarity, transparency, and completeness of the project documentation.

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11. Bibliography

The development of the Career Compass system involved referring to various online resources, documentation, and learning materials. These sources provided valuable guidance in understanding technologies, tools, and concepts used throughout the project.

11.1 Books and Study Materials

- Pressman, R. S. – *Software Engineering: A Practitioner's Approach*
- Sommerville, Ian – *Software Engineering*
- Korth, H. F. – *Database System Concepts*

11.2 Articles and Research Papers on Automation and Workflow

- "Introduction to Workflow Automation" – Make.com Blog
- "The Future of Automation in Business Processes" – McKinsey & Company
- "Workflow Automation: Benefits and Best Practices" – IBM Research
- "Understanding Webhooks and API Integration" – MDN Web Docs
- "Automation in Modern Web Applications" – GeeksforGeeks Articles

11.3 Websites and Online Resources

- https://www.w3schools.com
- https://developer.mozilla.org
- https://stackoverflow.com
- https://www.geeksforgeeks.org

These platforms were used to understand frontend development concepts, JavaScript validation, and general programming techniques.

11.4 Tools and Technologies Documentation

- Make.com Official Documentation – https://www.make.com
- Google Sheets Documentation – https://support.google.com/sheets
- HTML, CSS, JavaScript Documentation – MDN Web Docs

These sources helped in implementing automation workflows, managing data, and designing the user interface.

11.5 Research and Reference Materials

- Articles and blogs related to career guidance systems
- Online tutorials on automation and web development
- Documentation on API integration and webhooks

11.6 Acknowledgement of Resources

All the above resources were used strictly for learning and reference purposes during the development of this project. Proper understanding and implementation were carried out to ensure originality and correctness of the work.

Conclusion of Bibliography

The references listed above played a significant role in the successful completion of the Career Compass project. They provided essential knowledge and support in understanding various technologies and concepts used throughout the development process.
